

Use this parameter to enable the feedback supervision function. Use the feedback supervision to make sure that the PID feedback value stays in the set limits.

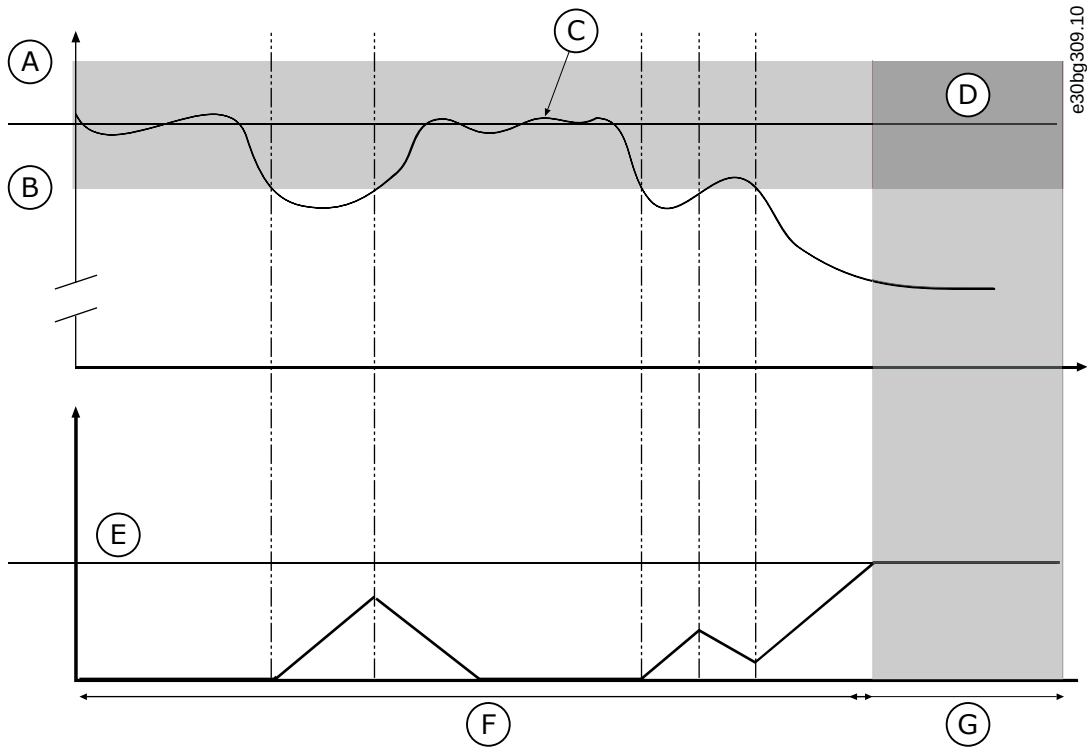


Illustration 82: The Feedback Supervision Function

A	Upper limit (ID 736)	E	Delay (ID 737)
B	Lower limit (ID 758)	F	Regulating mode
C	Actual value	G	Alarm or fault
D	Reference		

11.15.6.2 (ID 736) Upper Limit

Location in the menu: **P3.13.6.2**

Use this parameter to set the high limit for the PID feedback signal. If the value of the PID feedback signal goes above this limit for longer than the set time, a feedback supervision fault occurs.

11.15.6.3 (ID 758) Lower Limit

Location in the menu: **P3.13.6.3**

Use this parameter to set the low limit for the PID feedback signal. If the value of the PID feedback signal goes below this limit for longer than the set time, a feedback supervision fault occurs.

Set the upper limit and the lower limit around the reference. When the actual value is less or more than the limits, a counter starts to count up. When the actual value is between the limits, the counter counts down. When the counter gets a value that is higher than the value of P3.13.6.4 Delay, an alarm, or a fault shows. It is possible to select the response with parameter P3.13.6.5 (Response to PID1 Supervision Fault).

11.15.6.4 (ID 737) Delay

Location in the menu: **P3.13.6.4**

Use this parameter to set the maximum time for the PID feedback signal to stay outside the supervision limits before the feedback supervision fault occurs. If the target value is not reached in this time, a fault or alarm shows.

11.15.6.5 (ID 749) Response to PID Supervision Fault

Location in the menu: **P3.13.6.5**

Use this parameter to select the response of the drive to a 'PID Supervision' fault. If the PID feedback value is not in the supervision limits for longer than the supervision delay, a PID supervision fault occurs.

11.15.7 Pressure Loss Compensation

When pressurizing a long pipe that has many outlets, the best position for the sensor is in the middle of the pipe (the position 2 in the figure). It is also possible to put the sensor directly after the pump. This gives the right pressure directly after the pump, but farther in the pipe, the pressure drops with the flow.

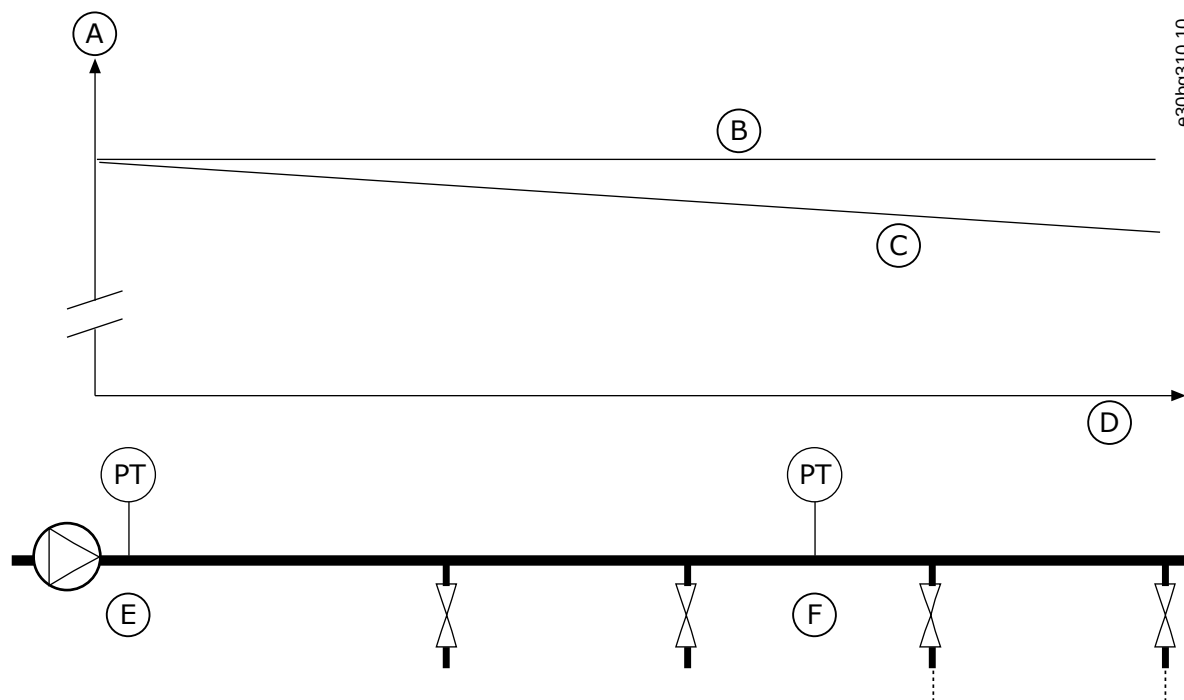


Illustration 83: The Position of the Pressure Sensor

A	Pressure	D	Pipe length
B	No flow	E	Position 1
C	With flow	F	Position 2

11.15.7.1 (ID 1189) Enable Setpoint 1

Location in the menu: **P3.13.7.1**

Use this parameter to enable the pressure loss compensation in the pump system. In a pressure-controlled system, this function compensates the pressure loss that occurs at the end of the pipe line due to the liquid flow.

11.15.7.2 (ID 1190) Setpoint 1 Max Compensation

Location in the menu: **P3.13.7.2**

Use this parameter to set the maximum compensation for PID setpoint value that is applied when the output frequency of the drive is at the maximum frequency. The compensation value is added to the actual setpoint value as a function of the output frequency. $\text{Setpoint compensation} = \text{max compensation} * (\text{FreqOut} - \text{MinFreq}) / (\text{MaxFreq} - \text{MinFreq})$.

The sensor is put in position 1. The pressure in the pipe stays constant when there is no flow. But with flow, the pressure decreases farther in the pipe. To compensate for this, lift the setpoint as the flow increases. Then the output frequency makes an estimate of the flow, and the setpoint increases linearly with the flow.

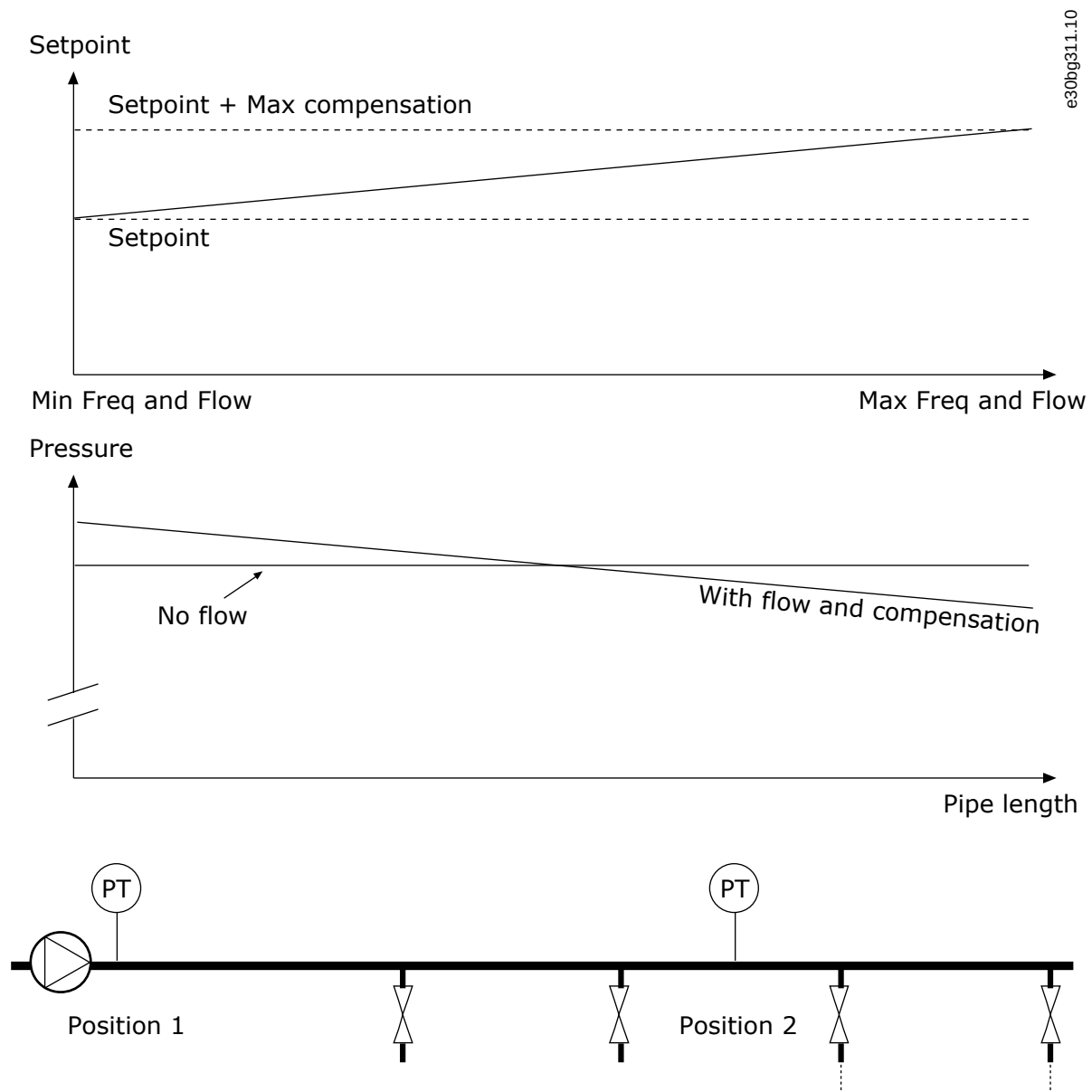


Illustration 84: Enable Setpoint 1 for Pressure Loss Compensation

11.15.8 Soft Fill

The Soft fill function is used to move the process to a set level at a slow speed before the PID controller starts to control. If the process does not go to the set level during the timeout, a fault shows.

It is possible to use the function to fill an empty pipe slowly and prevent strong currents of water that could break the pipe.

We recommend always using the Soft fill function when using the Multi-pump function.

11.15.8.1 (ID 1094) Soft Fill Function

Location in the menu: **P3.13.8.1**

Use this parameter to enable the Soft Fill function. It is possible to use the function to fill an empty pipe slowly and prevent strong currents of fluid that could break the pipe.

Selection number	Selection name	Description
0	Disabled	
1	Enabled (Level)	The drive operates at a constant frequency (P3.13.8.2 Soft Fill Frequency) until the PID feedback signal goes to the soft fill level (P3.13.8.3 Soft Fill Level). The PID controller starts the regulation. In addition, if the PID feedback signal does not go to the soft fill level in the soft fill timeout (P3.13.8.4 Soft Fill Timeout), a soft fill fault is shown (P3.13.8.4 Soft Fill Timeout is set to greater than 0). The soft fill mode is used in vertical installations.
2	Enabled (Timeout)	The drive operates at a constant frequency (P3.13.8.2 Soft Fill Frequency) until the soft fill time (P3.13.8.4 Soft Fill Timeout) goes. After the soft fill time, the PID controller starts the regulation. In this mode, the soft fill fault is not available. The soft fill mode is used in horizontal installations.

11.15.8.2 (ID 1055) Soft Fill Frequency

Location in the menu: **P3.13.8.2**

Use this parameter to set the frequency reference of the drive when the Soft Fill function is used.

11.15.8.3 (ID 1095) Soft Fill Level

Location in the menu: **P3.13.8.3**

Use this parameter to set the level below which the soft fill control is enabled when starting the drive. The drive operates at the PID start frequency until the feedback reaches the set value. Then the PID controller starts to control the drive. This parameter is applied if the soft fill function is set to 'Enabled, Level'.

11.15.8.4 (ID 1096) Soft Fill Timeout

Location in the menu: **P3.13.8.4**

Use this parameter to set the timeout time for the Soft Fill function. When the soft fill function is set to 'Enabled, Level', this parameter gives the timeout for the soft fill level, after which the soft fill fault occurs. When the soft fill function is set to 'Enabled, Timeout', the drive operates at the soft fill frequency until the time set by this parameter expires.

If the option Enabled (Timeout) is selected in parameter P3.13.8.1 Soft Fill Function, parameter Soft Fill Timeout gives the quantity of time that the drive operates at the constant soft fill frequency (P3.13.8.2 Soft Fill Frequency) before the PID controller starts the regulation.

11.15.8.5 (ID 748) Soft Fill Fault

Location in the menu: **P3.13.8.5**

Use this parameter to select the response of the drive to a 'PID Soft Fill' fault. If the PIDFeedback value does not reach the set level in the time limit, a soft fill fault occurs.

Selection number	Selection name	Description
0	No action	
1	Alarm	
2	Fault (stop according to stop mode)	
3	Fault (stop by coasting)	

11.15.9 Input Pressure Supervision

Use the Input pressure supervision to make sure that there is enough water in the inlet of the pump. When there is enough water, the pump does not suck air and there is no suction cavitation. To use the function, install a pressure sensor on the pump inlet.

If the input pressure of the pump goes below the set alarm limit, an alarm shows. The setpoint value of the PID controller decreases and causes the output pressure of the pump to decrease. If the pressure goes below the fault limit, the pump is stops and a fault shows.

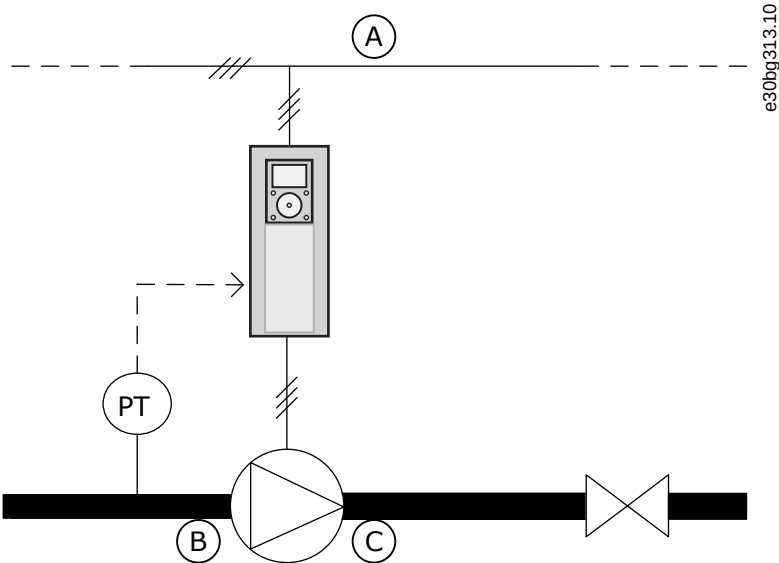


Illustration 85: The Location of the Pressure Sensor

A	Mains	C	Outlet
B	Inlet		

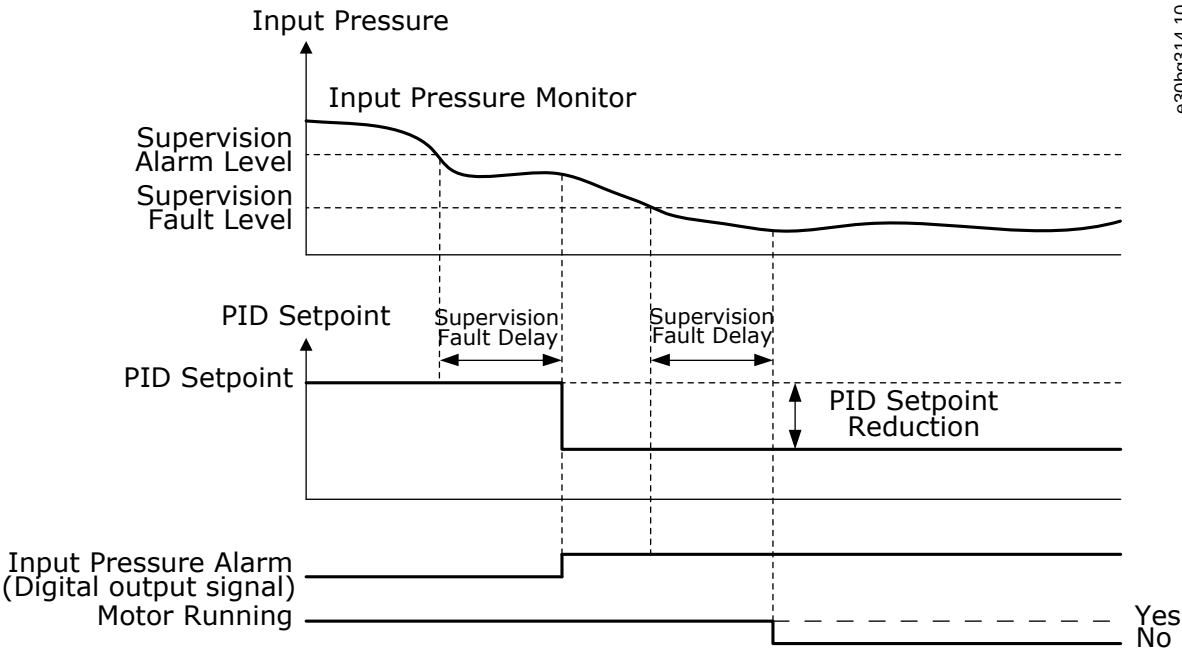


Illustration 86: The Input Pressure Supervision Function

11.15.9.1 (ID 1685) Enable Supervision

Location in the menu: **P3.13.9.1**

Use this parameter to enable the Input Pressure Supervision function. Use this function to make sure that there is enough fluid in the inlet of the pump.

11.15.9.2 (ID 1686) Supervision Signal

Location in the menu: **P3.13.9.2**

Use this parameter to select the source of the input pressure signal.

11.15.9.3 (ID 1687) Supervision Unit Selection

Location in the menu: **P3.13.9.3**

Use this parameter to select the unit for the input pressure signal. It is possible to scale the supervision signal (P3.13.9.2) to process units on the panel.

11.15.9.4 (ID 1688) Supervision Unit Decimals

Location in the menu: **P3.13.9.4**

Use this parameter to set the number of decimals for the input pressure signal unit. It is possible to scale the supervision signal (P3.13.9.2) to process units on the panel.

11.15.9.5 (ID 1689) Supervision Unit Minimum Value

Location in the menu: **P3.13.9.5**

Use this parameter to set the minimum value of the input pressure signal. Enter the value in the selected process unit. For example, an analog signal of 4...20 mA corresponds to the pressure of 0...10 bar.

11.15.9.6 (ID 1690) Supervision Unit Maximum Value

Location in the menu: **P3.13.9.6**

Use this parameter to set the maximum value of the input pressure signal. Enter the value in the selected process unit. For example, an analog signal of 4...20 mA corresponds to the pressure of 0...10 bar.

11.15.9.7 (ID 1691) Supervision Alarm Level

Location in the menu: **P3.13.9.7**

Use this parameter to set the limit for the input pressure alarm. If the measured input pressure goes below this limit, an input pressure alarm occurs.

11.15.9.8 (ID 1692) Supervision Fault Level

Location in the menu: **P3.13.9.8**

Use this parameter to set the limit for the input pressure fault. If the measured input pressure stays below this level for longer than the set time, an input pressure fault occurs.

11.15.9.9 (ID 1693) Supervision Fault Delay

Location in the menu: **P3.13.9.9**

Use this parameter to set the maximum duration for the input pressure to stay below the fault limit before an input pressure fault occurs.

11.15.9.10 (ID 1694) PID Setpoint Reduction

Location in the menu: **P3.13.9.10**

Use this parameter to set the rate of the reduction of the PID setpoint value when the measured input pressure is below the alarm limit.

11.15.10 Sleep Function When No Demand Is Detected

This function makes sure that the pump does not operate at a high speed when there is no demand in the system.

The function becomes active when the PID feedback signal and the output frequency of the drive stay in the specified hysteresis areas for longer than set with parameter P3.13.10.4 SNDD Supervision Time.

There are different hysteresis settings for the PID feedback signal and the output frequency. The hysteresis for the PID feedback (SNDD Error Hysteresis P3.13.10.2) is given in the selected process units around the PID setpoint value.

When the function is active, a short-time bias value (SNDD Actual Add) is added internally to the feedback value.

- If there is no demand in the system, the PID output, and the output frequency of the drive decrease to the direction of 0. If the PID feedback value stays in the hysteresis area, the drive goes to the Sleep mode.
- If the PID feedback value does not stay in the hysteresis area, the function is deactivated and the drive continues to operate.

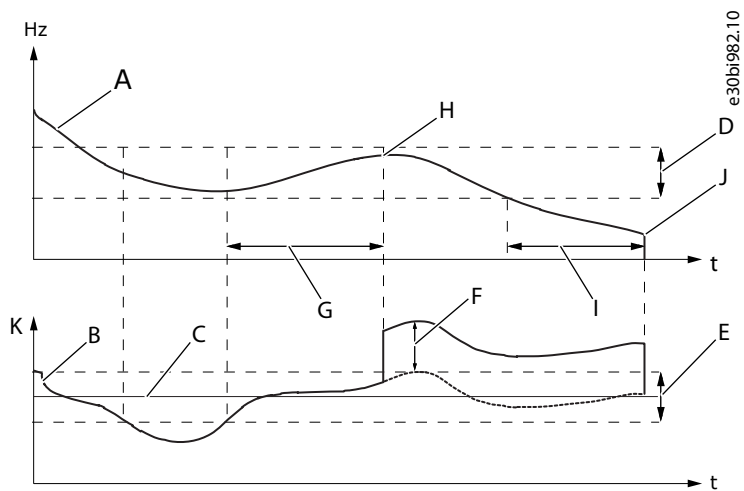


Illustration 87: Sleep, No Demand Detected

A	The output frequency of the drive	G	SNDD Supervision Time (P3.13.10.4)
B	The PID feedback value	H	The PID feedback value and the output frequency of the drive are in the hysteresis areas for the set time (SNDD Supervision Time). A bias value (SNDD Actual Add) is added to the PID feedback value.
C	The PID setpoint value	I	SP1 Sleep Delay Time (P3.13.5.2)
D	SNDD Frequency Hysteresis (P3.13.10.3)	J	The drive goes to the Sleep mode.
E	SNDD Error Hysteresis (P3.13.10.2) The hysteresis area around the PID setpoint value.	K	Process Unit (P3.13.1.4)
F	SNDD Actual Add (P3.13.10.5)		

11.15.10.1 (ID 1649) Sleep No Demand Detection Enable

Location in the menu: **P3.13.10.1**

Use this parameter to activate the Sleep No Demand Detection (SNDD) function.

11.15.10.2 (ID 1658) SNDD Error Hysteresis

Location in the menu: **P3.13.10.2**

Use this parameter to set the hysteresis for the error value of the PID controller.

11.15.10.3 (ID 1663) SNDD Frequency Hysteresis

Location in the menu: **P3.13.10.3**

Use this parameter to set the hysteresis for the output frequency of the drive.

11.15.10.4 (ID 1668) SNDD Supervision Time

Location in the menu: **P3.13.10.4**

Use this parameter to set the time that the output frequency of the drive and the error value of the PID controller must stay in the hysteresis areas before the SNDD function becomes active.

11.15.10.5 (ID 1669) SNDD Actual Add

Location in the menu: **P3.13.10.5**

Use this parameter to set the value that is added to the actual value of the PID feedback for a short time when the SNDD function is active.

11.15.11 Multi-setpoint

11.15.11.1 (ID 15560) Multi-setpoint 0

Location in the menu: **P3.13.12.1**

Use this parameter to set the preset PID setpoint value when the Multi-setpoint function is used. Make the selection of the multi-setpoint with the digital input signals.

11.15.11.2 (ID 15561) Multi-setpoint 1

Location in the menu: **P3.13.12.2**

Use this parameter to set the preset PID setpoint value when the Multi-setpoint function is used. Make the selection of the multi-setpoint with the digital input signals.

11.15.11.3 (ID 15562) Multi-setpoint 2

Location in the menu: **P3.13.12.3**

Use this parameter to set the preset PID setpoint value when the Multi-setpoint function is used. Make the selection of the multi-setpoint with the digital input signals.

11.15.11.4 (ID 15563) Multi-setpoint 3

Location in the menu: **P3.13.12.4**

Use this parameter to set the preset PID setpoint value when the Multi-setpoint function is used. Make the selection of the multi-setpoint with the digital input signals.

11.15.11.5 (ID 15564) Multi-setpoint 4

Location in the menu: **P3.13.12.5**

Use this parameter to set the preset PID setpoint value when the Multi-setpoint function is used. Make the selection of the multi-setpoint with the digital input signals.

11.15.11.6 (ID 15565) Multi-setpoint 5

Location in the menu: **P3.13.12.6**

Use this parameter to set the preset PID setpoint value when the Multi-setpoint function is used. Make the selection of the multi-setpoint with the digital input signals.

11.15.11.7 (ID 15566) Multi-setpoint 6

Location in the menu: **P3.13.12.7**

Use this parameter to set the preset PID setpoint value when the Multi-setpoint function is used. Make the selection of the multi-setpoint with the digital input signals.

11.15.11.8 (ID 15567) Multi-setpoint 7

Location in the menu: **P3.13.12.8**

Use this parameter to set the preset PID setpoint value when the Multi-setpoint function is used. Make the selection of the multi-setpoint with the digital input signals.

11.15.11.9 (ID 15569) Multi-setpoint 9

Location in the menu: **P3.13.12.10**

Use this parameter to set the preset PID setpoint value when the Multi-setpoint function is used. Make the selection of the multi-setpoint with the digital input signals.

11.15.11.10 (ID 15570) Multi-setpoint 10

Location in the menu: **P3.13.12.11**

Use this parameter to set the preset PID setpoint value when the Multi-setpoint function is used. Make the selection of the multi-setpoint with the digital input signals.

11.15.11.11 (ID 15571) Multi-setpoint 11

Location in the menu: **P3.13.12.11**

Use this parameter to set the preset PID setpoint value when the Multi-setpoint function is used. Make the selection of the multi-setpoint with the digital input signals.

11.15.11.12 (ID 15572) Multi-setpoint 12

Location in the menu: **P3.13.12.13**

Use this parameter to set the preset PID setpoint value when the Multi-setpoint function is used. Make the selection of the multi-setpoint with the digital input signals.

11.15.11.13 (ID 15573) Multi-setpoint 13

Location in the menu: **P3.13.12.13**

Use this parameter to set the preset PID setpoint value when the Multi-setpoint function is used. Make the selection of the multi-setpoint with the digital input signals.

11.15.11.14 (ID 15574) Multi-setpoint 14

Location in the menu: **P3.13.12.15**

Use this parameter to set the preset PID setpoint value when the Multi-setpoint function is used. Make the selection of the multi-setpoint with the digital input signals.

11.15.11.15 (ID 15575) Multi-setpoint 15

Location in the menu: **P3.13.12.16**

Use this parameter to set the preset PID setpoint value when the Multi-setpoint function is used. Make the selection of the multi-setpoint with the digital input signals.

11.15.11.16 (ID 15576) Multi-setpoint Selection 0

Location in the menu: **P3.13.12.17**

Use this parameter to select the digital input signal that is used as a selector for the PID Multi-setpoint function.

11.15.11.17 (ID 15577) Multi-setpoint Selection 1

Location in the menu: **P3.13.12.18**

Use this parameter to select the digital input signal that is used as a selector for the PID Multi-setpoint function.

11.15.11.18 (ID 15578) Multi-setpoint Selection 2

Location in the menu: **P3.13.12.19**

Use this parameter to select the digital input signal that is used as a selector for the PID Multi-setpoint function.

11.15.11.19 (ID 15579) Multi-setpoint Selection 3

Location in the menu: **P3.13.12.20**

Use this parameter to select the digital input signal that is used as a selector for the PID Multi-setpoint function.

11.16 External PID Controller

11.16.1 (ID 1630) Enable External PID

Location in the menu: **P3.14.1.1**

Use this parameter to enable the PID controller.

This controller is for external use only. It can be used with an analog output.

11.16.2 (ID 1049) Start Signal

Location in the menu: **P3.14.1.2**

Use this parameter to set the signal for starting and stopping the PID controller 2 for external usage.

If the PID2 controller is not enabled in the Basic menu for PID2, this parameter has no effect.

11.17 Multi-pump Function

11.17.1 The Multi-pump Function

The Multi-pump function allows controlling a system where the maximum of 8 motors, for example, pumps, fans or compressors operate in parallel. The internal PID controller of the drive operates the necessary quantity of motors and controls the speed of the motors, when there is demand.

11.17.1.1 Commissioning Multi-pump (Multidrive)

The check list helps in the configuration of the basic settings of the Multi-pump (multidrive) system. If the keypad is used for parameterization, the application wizard helps to make the basic settings. Start the commissioning with the drives that have the PID feedback signal (pressure sensor, for example) connected to an analog input (default: AI2). Go through all the drives in the system.

Procedure

1. Examine the wiring.

- See the correct power cabling (mains cable, motor cable) of the drive in Installation Guide.
- See the correct control cabling (I/O, PID feedback sensor, communication) in [Illustration 20](#) and in [Illustration 18](#).
- If redundancy is necessary, make sure that the PID feedback signal (by default: AI2) is connected to a minimum of 2 drives. See the wiring instructions in [Illustration 20](#).

2. Do a power-up of the drive and start the parameterization.

- Start the parameterization with the drives that have the PID feedback signal connected. These drives can operate as the master of the Multi-pump system.
- Parameterization can be done with the keypad or PC the tool.

3. Select the Multi-pump (multidrive) application configuration with parameter P1.2.

- Most of the Multi-pump-related settings and configurations are made automatically, when the Multi-pump (multidrive) application is selected with parameter P1.2 Application (ID 212). See [4.1.5 Multi-pump \(Multidrive\) Application Wizard](#).
- If the keypad is used for the parameterization, the Application wizard starts when parameter P1.2 Application (ID 212) is changed. The Application wizard helps with Multipump-related questions.

4. Set the motor parameters.

- Set the motor nameplate parameters specified by the nameplate of the motor.

5. Set the total number of drives used in the Multi-pump system.

- This value is set with parameter P1.35.14 Quick Setup Parameter Menu.
- The same parameter is in the menu Parameters -> Group 3.15 -> P3.15.2.
- By default, the Multi-pump system has 3 pumps (drives).

6. Select the signals that are connected to the drive.

- Go to parameter P1.35.16 (Quick Setup Parameter Menu).
- The same parameter is in the menu Parameters -> Group 3.15 -> P3.15.4.
- If the PID feedback signal is connected, the drive can operate as the master of the Multi-pump system. If the signal is not connected, the drive operates as a slave unit.
- Select Signals connected, if the start and the PID feedback signals (the pressure sensor, for example) are connected to the drive.
- Select Start signal only, if only the start signal is connected to the drive (the PID feedback signal is not connected).
- Select Not connected, if the start or the PID feedback signals are not connected to the drive.

7. Set the ID number of the pump.

- Go to parameter P1.35.15 (Quick Setup Parameter Menu).
- The same parameter is in the menu Parameters -> Group 3.15 -> P3.15.3.
- Each drive in the Multi-pump system must have an ID number that no other drive has for the correct communication between drives. The ID numbers must be in a numerical order and start from number 1.
- The drives, which have a PID feedback signal connected, have the smallest ID numbers (for example, ID 1 and ID 2). It gives the shortest possible start-up delay when doing a power-up of the system.

8. Configure the Interlock function.

- Go to parameter P1.35.17 (Quick Setup Parameter Menu).
- The same parameter is in menu Parameters -> Group 3.15 -> P3.15.5.
- By default, the interlock function is disabled.
- Select Enabled, if the interlock signal is connected to the digital input DI5 of the drive. The interlock signal is the digital input signal that tells if this pump is available in the Multi-pump system.
- Select Not Used, if the interlock signal is not connected to the digital input DI5 of the drive. The system sees that all the pumps in the Multi-pump system are available.

9. Examine the source of the PID setpoint signal.

- By default, the PID setpoint value comes from parameter P1.35.9 Keypad Setpoint 1.
- If necessary, change the source of the PID setpoint signal with parameter P1.35.8. Select the analog input or Fieldbus Process Data In 1–8, for example.

The basic settings of the Multi-pump system are completed. Use the check list also when configuring the next drives in the system.

11.17.2 System Configuration

11.17.2.1 Single Drive Configuration

The Single drive mode controls a system of 1 variable speed pump and the maximum of 7 auxiliary pumps. The internal PID controller of the drive controls the speed of 1 pump and gives control signals with relay outputs to start or stop the auxiliary pumps. External contactors are necessary for the switch of the auxiliary pumps to the mains.

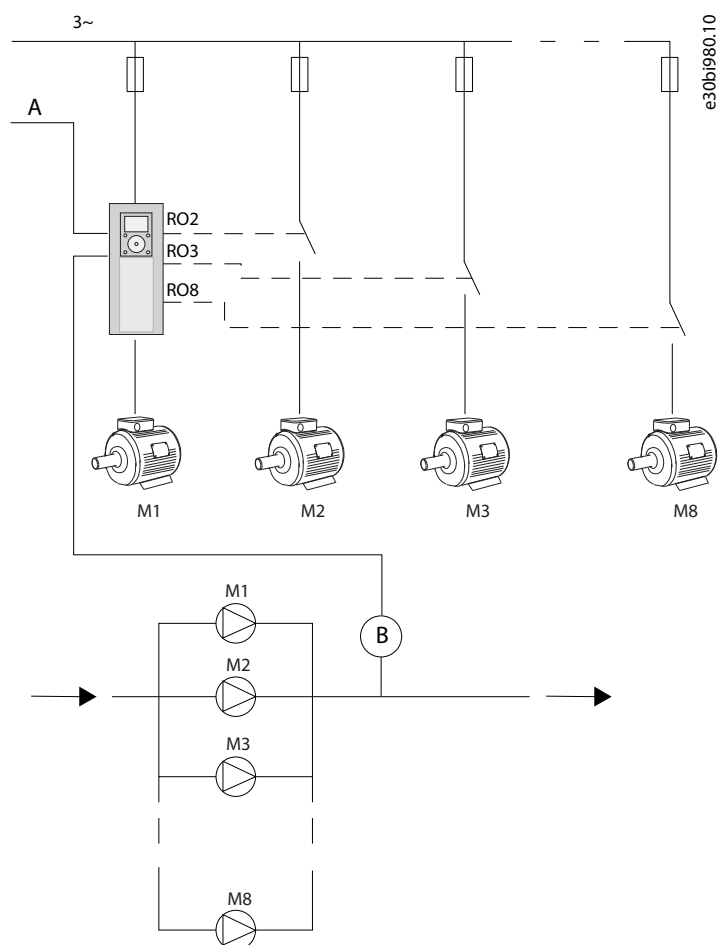


Illustration 88: Single Drive Configuration

A	Start / Stop
B	Pressure sensor

11.17.2.2 Multidrive Configuration

The Multidrive modes (Multimaster and Multifollower) control a system that has the maximum 8 variable speed pumps. Each pump is controlled by a drive. The internal PID controller of the drive controls all pumps. The drives use a communication bus (Modbus RTU) for communication.

The figure shows the Multidrive configuration principle. See also the general electric diagram of a Multi-pump system in [Illustration 20](#).

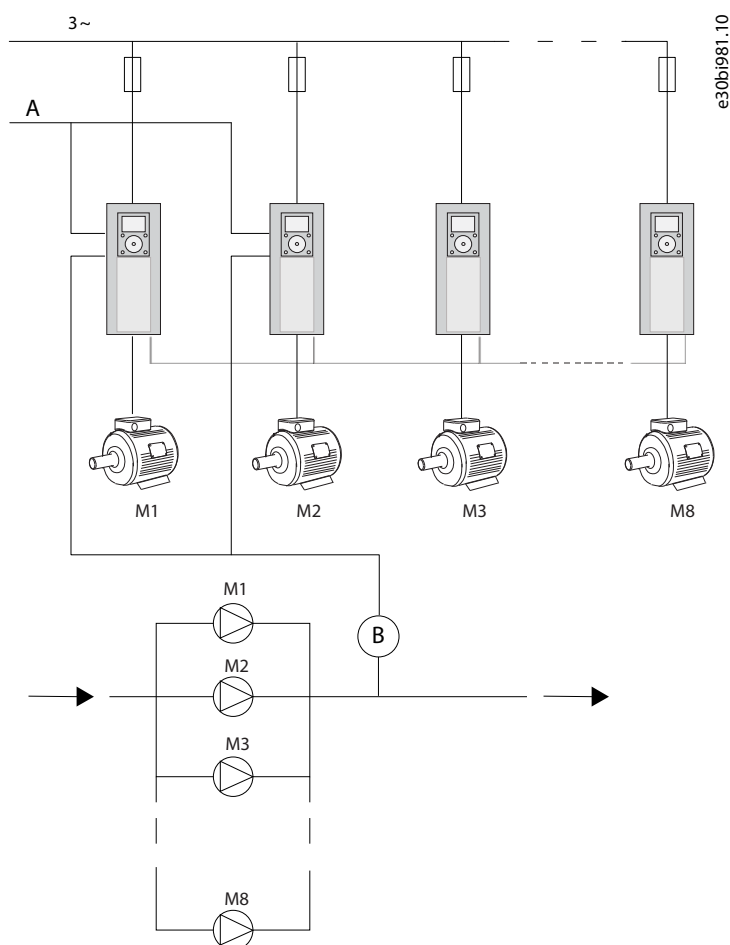


Illustration 89: Multidrive Configuration

A	Start / Stop
B	Pressure sensor

11.17.2.3 (ID 1785) Multi-pump Mode

Location in the menu: **P3.15.1**

Use this parameter to select the configuration and the control mode of the Multi-pump system. The Multi-pump function allows controlling a maximum of 8 motors (that is, pumps, fans, compressors) with the PID control.

0 = Single drive

The Single drive mode controls a system that has 1 pump that can change speed and the maximum of 7 auxiliary pumps. The internal PID controller of the drive controls the speed of 1 pump and gives control signals with relay outputs to start or stop the auxiliary pumps.

External contactors are necessary for the switch of the auxiliary pumps to the mains. 1 of the pumps is connected to the drive and controls the system. When the pump in control sees that it is necessary to have more capacity (operates at the maximum frequency), the drive gives the control signal with the relay output to start the next auxiliary pump. When the auxiliary pump starts, the pump in control continues to control and starts from the minimum frequency.

When the pump that controls the system sees that there is too much capacity (operates at the minimum frequency), the pump makes the started auxiliary pump to stop. If no auxiliary pumps operate when the pump in control sees the overcapacity, the pump goes to the Sleep mode (if the Sleep function is enabled).

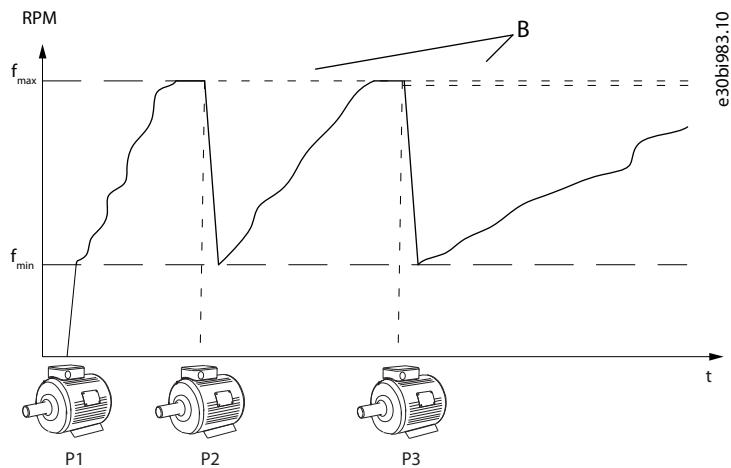


Illustration 90: Control in the Single Drive Mode

P1	The pump that controls the system
B	The auxiliary pumps connected to the mains (direct on-line)

1 = Multifollower

The Multifollower mode controls a system that has the maximum 8 pumps that can change speed. Each pump is controlled by a drive. The internal PID controller of the drive controls all the pumps.

1 of the pumps always controls the system. When the pump in control sees that it is necessary to have more capacity (operates at the maximum frequency), the pump uses the communication bus to make the next pump to start. The next pump increases speed and starts to operate at the speed of the pump in control. Auxiliary pumps operate at the speed of the pump that controls the system.

When the pump that controls the system sees that there is too much capacity (operates at the minimum frequency), it makes the started pump to stop. If no auxiliary pumps operate when the pump in control sees overcapacity, the pump goes to the Sleep mode (if the Sleep function is enabled).

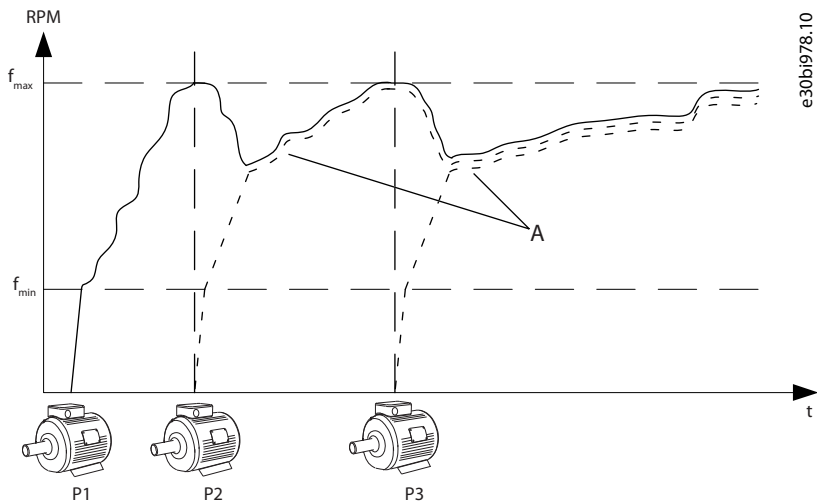


Illustration 91: Control in the Multifollower Mode

P1	The pump controls the system.	P3	The pump follows the speed of P1.
P2	The pump follows the speed of P1.	A	Curve A shows the auxiliary pumps that follow the speed of pump 1.

2 = Multimaster

The Multimaster mode controls a system that has the maximum 8 pumps that can change speed. Each pump is controlled by a drive. The internal PID controller of the drive controls all the pumps.

1 of the pumps always controls the system. When the pump in control sees that it is necessary to have more capacity (operates at the maximum frequency), it locks to a constant production speed and makes the next pump to start and to control the system. When the pump that controls the system sees that there is too much capacity (operates at the minimum frequency), it stops. The pump that operates at a constant production speed starts to control the system. If there are many pumps that operate at a constant production speed when the pump in control sees the overcapacity, the pump goes to the Sleep mode (if the Sleep function is enabled).

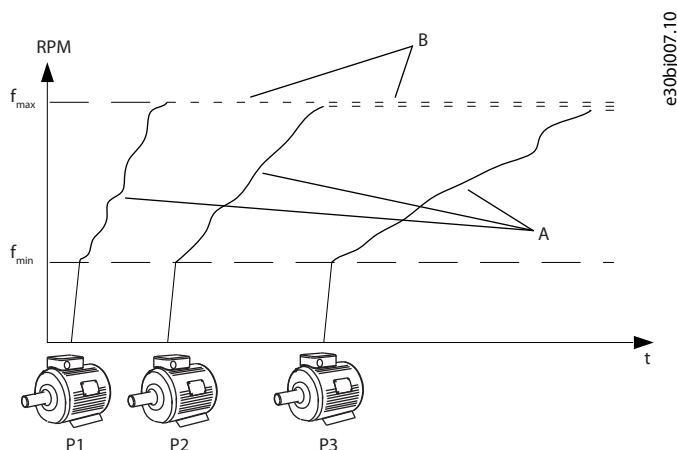


Illustration 92: Control in the Multimaster Mode

- | | |
|---|---|
| A | Curves A show the control of the pumps |
| B | The pumps are locked to the constant production frequency |

11.17.2.4 (ID 1001) Number of Pumps

Location in the menu: **P3.15.2**

Use this parameter to set the total number of motors/pumps used in the Multi-pump system. The maximum number of pumps in the Multi-pump system is 8.

Set this parameter in the installation. If 1 drive is removed to do the servicing of the pump, for example, it is not necessary to change this parameter.

In the Multifollower and Multimaster modes, all drives must have the same value in this parameter for the correct communication between the drives.

11.17.2.5 (ID 1500) Pump ID Number

Location in the menu: **P3.15.3**

Use this parameter to set the ID number of the drive. This parameter is used only in the Multifollower and Multimaster modes.

Each drive in the Multi-pump system must have a unique sequence (ID) number, always start from 1.

Pump number 1 is always the primary master of the Multi-pump system. Drive number 1 controls the process and the PID controller. The PID feedback and the PID setpoint signals must be connected to the drive number 1.

If the drive number 1 is not available in the system, there is a power-down of the drive, for example, the next drive starts to operate as a secondary master of the Multi-pump system.

The communication between the drives is not correct, if:

- the Pump ID numbers are not in a numerical order (start from 1), or
- 2 drives have the same ID number.

11.17.2.6 (ID 1782) Start and Feedback Signals

Location in the menu: **P3.15.4**

Use this parameter to select the signals that are connected to the drive.

0 = The start and the PID feedback signals not connected to the drive in question

1 = Only the start signals connected to the drive in question

2 = The start and the PID feedback signals connected to the drive in question

This parameter specifies the operation mode (master or slave) in the Multi-pump system. The drives that have the start command and the PID feedback signals connected, can operate as the master drive in the Multi-pump system. If there are many drives in the Multi-pump system that have all the signals connected, the drive with the lowest Pump ID Number (P3.15.3) starts to operate as the master.

11.17.3 Interlocks

The interlocks tell the Multi-pump system that a motor is not available. It can occur when the motor is removed from the system for maintenance or bypassed for manual control.

11.17.3.1 (ID 1032) Pump Interlocking

Location in the menu: **P3.15.5**

Use this parameter to enable or disable the interlocks. Interlock signal tells to the Multi-pump system if the motor is available or not. Interlock signals are given with DI signals. To use the interlocks, enable the parameter P3.15.2. Select the status for each motor with a digital input (the parameters from P3.5.1.34 to P3.5.1.39). If the value of the input is CLOSED, that is, active, the Multi-pump logic connects the motor to the Multi-pump system.

11.17.4 Feedback Sensor Connection in a Multi-pump System

To get the best precision and redundancy in the Multi-pump system, use the feedback sensors for each drive.

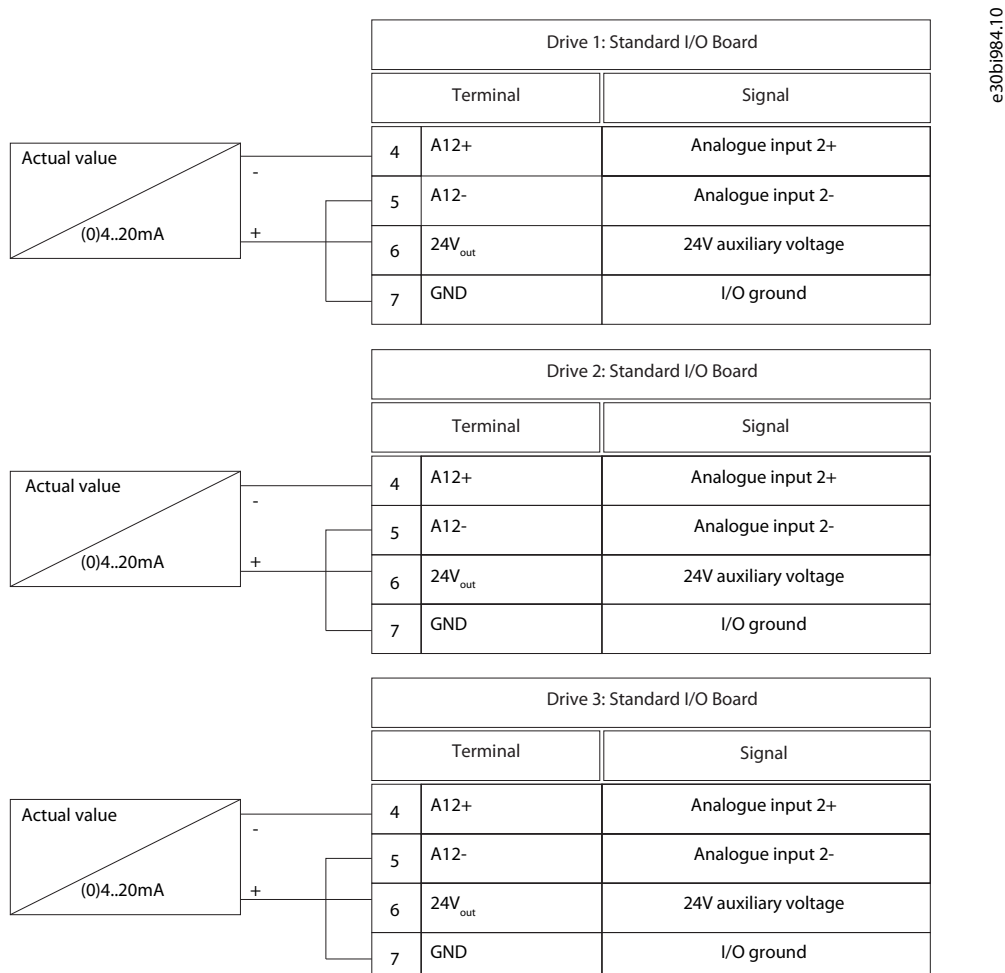


Illustration 93: Wiring of the Feedback Sensors for Each Drive

Same sensors can also be used for all the drives. The sensor (transducer) can be supplied by a 24 V external supply or from the control board the drive.

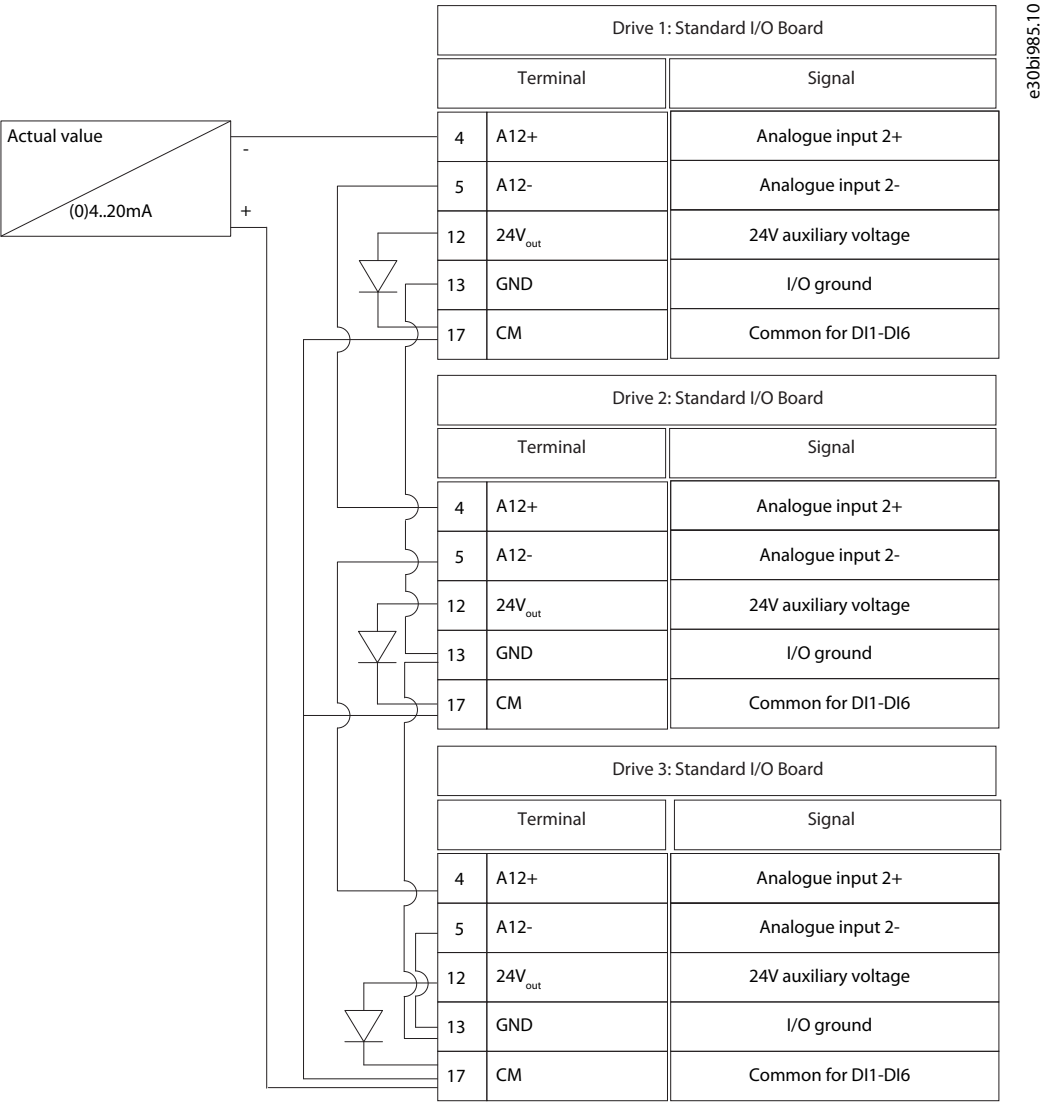


Illustration 94: Wiring of the Same Sensor for All Drives (Supplied from the I/O Board of the Drive)

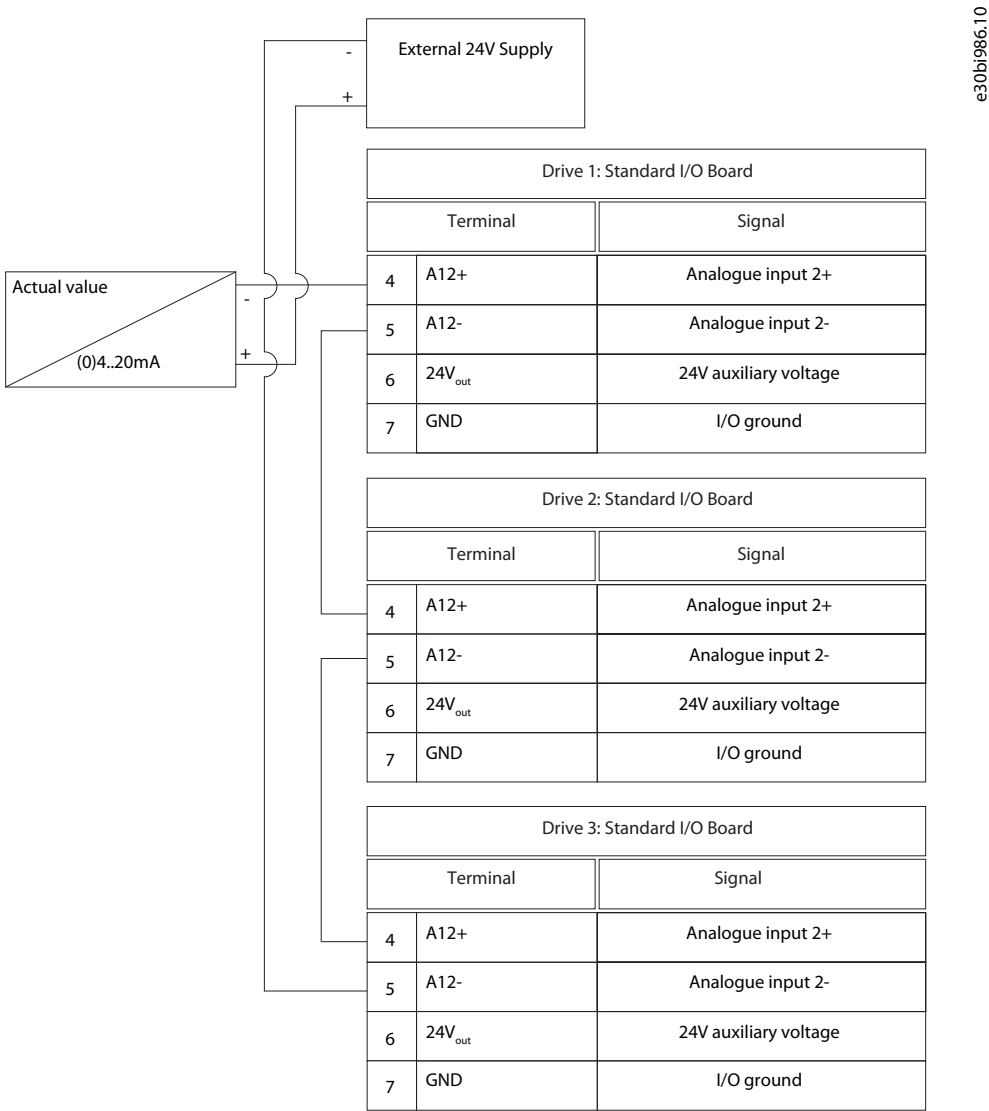


Illustration 95: Wiring of the Same Sensor for All Drives (Supplied from an External 24 V)

If a sensor is supplied from the I/O board of the drive and the diodes are connected between terminals 12 and 17, the digital inputs must be isolated from the ground. Set the isolation DIP switch to *Float*.
The digital inputs are active when they are connected to GND, which is the default condition.

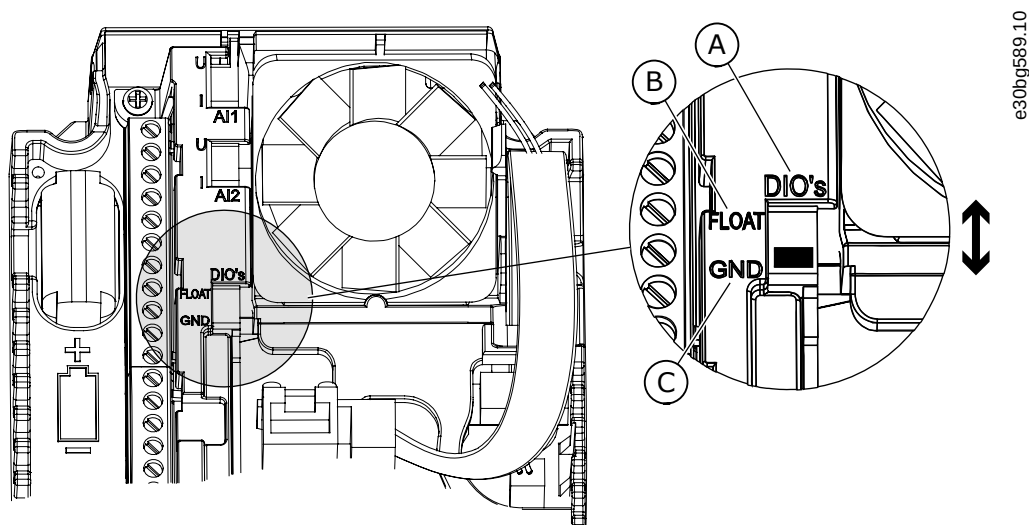


Illustration 96: Isolation DIP Switch

A	Digital inputs	C	Connected to GND (default)
B	Float		

11.17.4.1 (ID 1027) Autochange

Location in the menu: **P3.15.6**

Use this parameter to enable or disable the rotation of the start sequence and the priority of motors. The autochange changes the sequence in which the motors start to wear the motors equally.

The monitoring values V2.10.7 Pump 1 Running Time – V2.10.14 Pump 8 Running Time count the run-time of the motors. The autochange function sets the start sequence according to the run-times. The motor that has the lowest running hours starts first and the motor that has the highest running hours starts last.

Selection number	Selection name	Description
0	Disabled	In normal operation, the sequence of the motors is always 1, 2, 3, 4, 5. The sequence can change during the operation if interlocks are added or removed. After the drive stops, the sequence always changes back.
1	Enabled (interval)	The system changes the sequence at intervals to wear the motors equally. The intervals of the autochange can be adjusted with parameter P3.15.8. The timer of the autochange interval operates only when the Multi-pump system operates.
2	Enabled (week-days)	The start sequence changes at the selected weekday and time of day. Make the selection with parameters P3.15.9 and P3.15.10. To use this mode, an RTC battery must be installed in the drive.

Example

After an autochange, the first motor is put last. The other motors move up 1 position.

The start sequence of the motors: 1, 2, 3, 4, 5

--> Autochange -->

The start sequence of the motors: 2, 3, 4, 5, 1

--> Autochange -->

The start sequence of the motors: 3, 4, 5, 1, 2

11.17.4.2 (ID 1028) Autochanged Pumps

Location in the menu: **P3.15.7**

Use this parameter to include the controlled motor/pump in the autochange and interlock system.

Selection number	Selection name	Description
0	Auxiliary pumps	The drive is always connected to Motor 1. The interlocks do not affect Motor 1. Motor 1 is not included in the autochange logic.
1	All pumps	It is possible to connect the drive to any of the motors in the system. The interlocks affect all the motors. All the motors are included in the autochange logic.

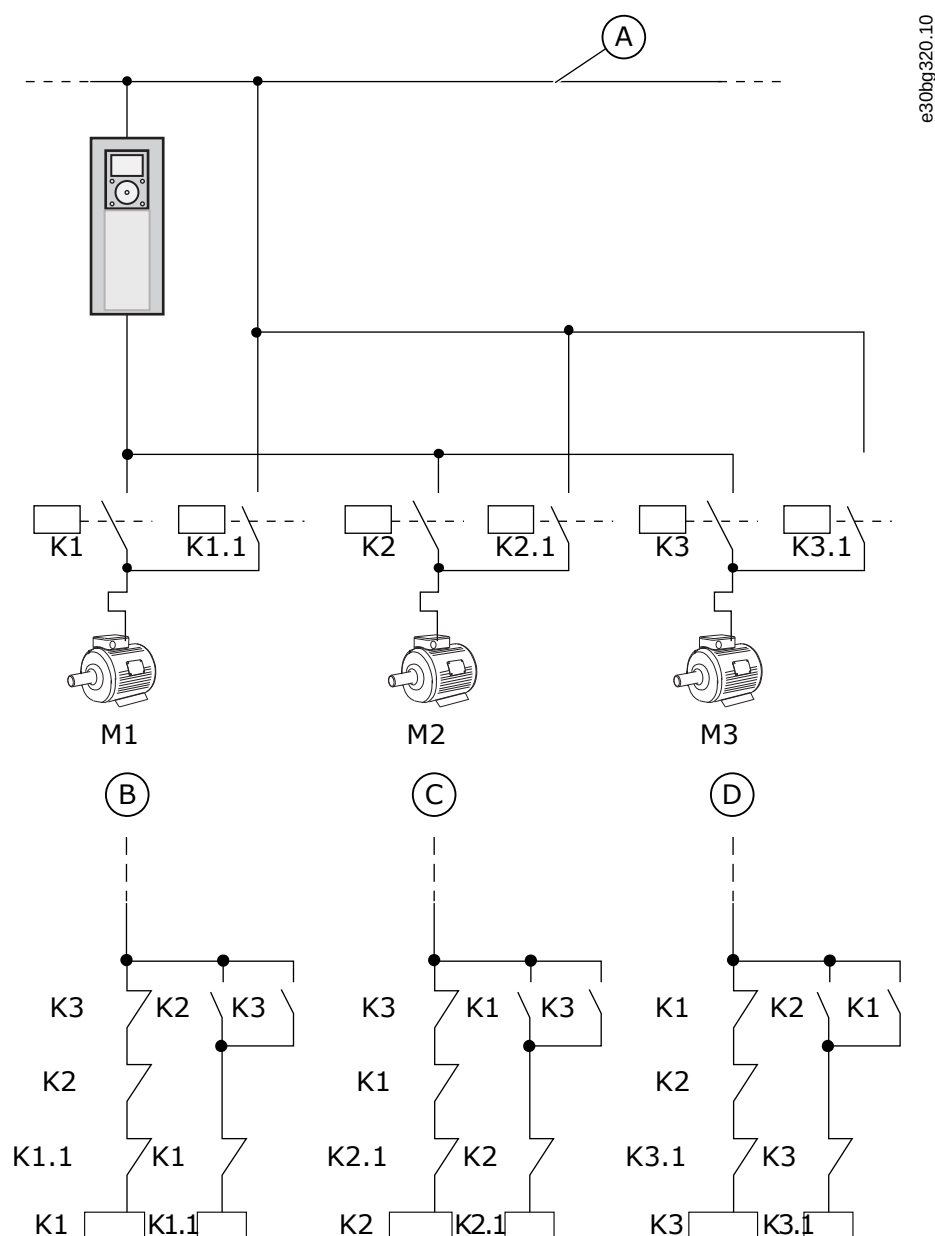
The connections are different for the parameter values 0 and 1.

Selection 0, Auxiliary pumps

The drive is directly connected to Motor 1. The other motors are auxiliary motors. They are connected to the mains by contactors, and controlled by relays of the drive. The autochange or the interlock logic does not affect Motor 1.

Selection 1, All pumps

To include the regulating motor in the autochange or in the interlock logic, obey the instructions in the following figure. 1 relay controls each motor. The contactor logic always connects the first motor to the drive, and the next motors to the mains.



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Illustration 97: Selection 1

A	Mains	C	Motor 2 control from relay
B	Motor 1 control from relay	D	Motor 3 control from relay

11.17.4.3 (ID 1029) Autochange Interval

Location in the menu: **P3.15.8**

Use this parameter to adjust the autochange intervals. The Autochange interval is the time after which the autochange function starts if the used capacity is below the set level. The value of this timer does not run when the Multi-pump system is stopped or in the sleep mode. To use the parameter, select Enabled (interval) with parameter P3.15.6 Autochange Mode.

The autochange occurs, if:

- the Multi-pump system operates (the start command is active)
- the autochange interval time goes
- the pump that controls the system operates below the frequency specified by parameter P3.15.11 Autochange Frequency Limit
- the number of pumps that operate is less or equal to the limit specified by parameter P3.15.12 Autochange Pump Limit.

11.17.4.4 (ID 1786) Autochange Days

Location in the menu: **P3.15.9**

Use this parameter to set the days of the week when the autochange function starts. The value of this parameter is applied if the autochange mode is 'Enabled (Weekdays)'.

11.17.4.5 (ID 1787) Autochange Time of Day

Location in the menu: **P3.15.10**

Use this parameter to set the time of the day when the autochange function starts. The value of this parameter is applied if the autochange mode is Enabled (Weekdays). To use the parameters, select Enabled (Weekdays) with parameter P3.15.6 Autochange.

The autochange occurs, if:

- the Multi-pump system operates (the start command is active)
- it is the autochange weekday and the time of day
- the pump that controls the system operates below the frequency specified by parameter P3.15.11 Autochange Frequency Limit
- the number of pumps that operate is less or equal to the limit specified by parameter P3.15.12 Autochange Pump Limit.

11.17.4.6 (ID 1031) Autochange Frequency Limit

Location in the menu: **P3.15.11**

Use this parameter to set the autochange frequency limit. The autochange frequency limit is the limit below which the output frequency of the regulating drive must stay for the autochange to start.

11.17.4.7 (ID 1030) Autochange Pump Limit

Location in the menu: **P3.15.12**

Use this parameter to set the amount of pumps used in the Multi-pump function. The Autochange Pump Limit is the limit below which the number of the running motors must stay for the autochange to start.

If the number of pumps that operate in the Multi-pump system is less or equal to the limit specified by parameter P3.15.12 and the pump that controls the system operates below the frequency specified by parameter P3.15.11, the autochange can occur.

These parameters are used in the Single drive mode, because the autochange can restart the system (depending on the quantity of motors that operate).

In the Multifollower and Multimaster modes, set these parameters to the maximum values to make it possible for the autochange to occur immediately at the autochange time. In the Multifollower and Multimaster modes, the quantity of pumps that operate does not affect the autochange.

11.17.4.8 (ID 1097) Bandwidth

Location in the menu: **P3.15.13**

Use this parameter to set the bandwidth area around the PID setpoint for starting and stopping of the auxiliary motors. When the PID feedback value stays in the bandwidth area, the auxiliary motors do not start or stop. The value of this parameter is given as a percentage of the setpoint.

11.17.4.9 (ID 1098) Bandwidth Delay

Location in the menu: **P3.15.14**

Use this parameter to set the duration before the auxiliary motors start or stop. When the PID feedback is not in the bandwidth area, the time that is set with this parameter must go before the auxiliary motors start or stop. The number of pumps that operate increases or decreases, if the PID controller cannot keep the process value (feedback) in the specified bandwidth around the setpoint.

The bandwidth area is specified as a percentage of the PID setpoint. When the PID feedback value stays in the bandwidth area, it is not necessary to increase or decrease the number of pumps that operate.

When the feedback value goes out of the bandwidth area, the quantity of time specified by parameter P3.15.14 must go before the number of pumps that operate increases or decreases. More pumps must be available.

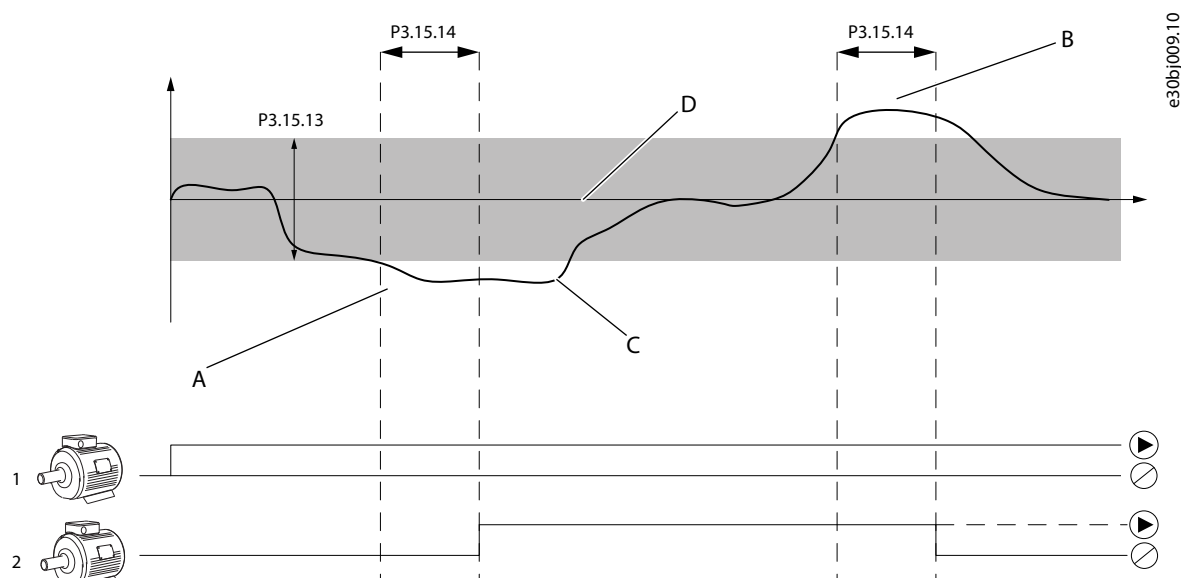


Illustration 98: The Start or Stop of the Auxiliary Pumps (P3.15.13 = Bandwidth, P3.15.14 = Bandwidth delay)

A The pump that controls the system operates at a frequency that is near the maximum (-2 Hz). It increases the number of pumps that operate. B The pump that controls the system operates at a frequency that is near the minimum (+2 Hz). It decreases the number of pumps that operate.	C The number pumps that operate increases or decreases, if the PID controller cannot keep the process value feedback in the specified bandwidth around the setpoint. D The specified bandwidth around the setpoint.
---	---

11.17.4.10 (ID 1513) Constant Production Speed

Location in the menu: **P3.15.15**

Use this parameter to set the constant speed at which the motor locks when the next motor starts in the Multimaster system. The value of this parameter is given as a percentage of the minimum frequency to the maximum frequency.

11.17.4.11 (ID 1187) Running Pumps Limit

Location in the menu: **P3.15.16**

Use this parameter to set the maximum number of motors that run at the same time in the Multi-pump system.

If the value of parameter P3.15.2 Number of Pumps changes, the same value changes automatically to this parameter.

Example

The Multi-pump system has 3 pumps, but only 2 pumps can operate at the same time. The third pump is installed in the system for redundancy. The number of pumps that can operate at the same time:

- Running Pump Limit = 2

11.17.4.12 (ID 426) Pump 1 Interlock

Location in the menu: **P3.15.17.1**

Use this parameter to select the digital input signal that is used as interlock signal for the multi-pump system.

When the Pump interlocking function (P3.15.5) is enabled, the drive reads the statuses of the digital inputs of the pump interlock (feedback). When the input is CLOSED, the motor is available for the Multi-pump system.

When the Pump interlocking function (P3.15.5) is disabled, the drive does not read the statuses of the digital inputs of the pump interlock (feedback). The Multi-pump system sees all pumps in the system as available.

- In the Single drive mode, the digital input signal that is selected with this parameter shows the interlock status of pump 1 in the Multi-pump system.
- In the Multifollower and Multimaster modes, the digital input signal that is selected with this parameter shows the interlock status of the pump that is connected to this drive.

11.17.4.13 (ID 427) Pump 2 Interlock

Location in the menu: **P3.15.17.2**

Use this parameter to select the digital input signal that is used as interlock signal for the multi-pump system.

11.17.4.14 (ID 428) Pump 3 Interlock

Location in the menu: **P3.15.17.3**

Use this parameter to select the digital input signal that is used as interlock signal for the multi-pump system.

11.17.4.15 (ID 429) Pump 4 Interlock

Location in the menu: **P3.15.17.4**

Use this parameter to select the digital input signal that is used as interlock signal for the multi-pump system.

11.17.4.16 (ID 430) Pump 5 Interlock

Location in the menu: **P3.15.17.5**

Use this parameter to select the digital input signal that is used as interlock signal for the multi-pump system.

11.17.4.17 (ID 486) Pump 6 Interlock

Location in the menu: **P3.15.17.6**

Use this parameter to select the digital input signal that is used as interlock signal for the multi-pump system.

11.17.4.18 (ID 487) Pump 7 Interlock

Location in the menu: **P3.15.17.7**

Use this parameter to select the digital input signal that is used as interlock signal for the multi-pump system.

11.17.4.19 (ID 488) Pump 8 Interlock

Location in the menu: **P3.15.17.8**

Use this parameter to select the digital input signal that is used as interlock signal for the multi-pump system.

These parameters are used in the Single drive mode only.

When the Pump interlocking function (P3.15.5) is enabled, the drive reads the statuses of the digital inputs of the pump interlock (feedback). When the input is CLOSED, the motor is available for the Multi-pump system.

When the Pump interlocking function (P3.15.5) is disabled, the drive does not read the statuses of the digital inputs of the pump interlock (feedback). The Multi-pump system sees all pumps in the system as available.

11.17.5 Overpressure Supervision

Use the Overpressure supervision function in a Multi-pump system. For example, when the primary valve of the pump system is closed quickly, the pressure in the pipe lines increases. The pressure can increase too quickly for the PID controller. To prevent that the pipes break, the overpressure supervision stops the auxiliary motors in the Multi-pump system.

11.17.5.1 (ID 1698) Enable Overpressure Supervision

Location in the menu: **P3.15.18.1**

Use this parameter to enable the Overpressure supervision.

The overpressure supervision monitors the feedback signal of the PID controller, that is, the pressure. If the signal becomes higher than the overpressure level, it stops all the auxiliary pumps immediately. Only the regulating motor continues to operate. When the pressure decreases, the system continues to operate, and connects the auxiliary motors again one at a time.

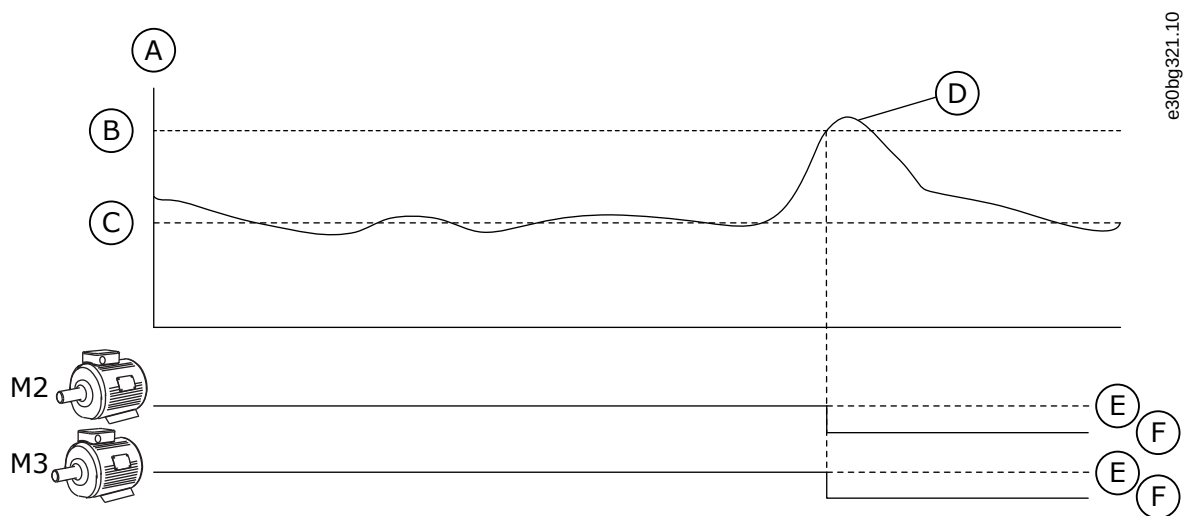


Illustration 99: The Overpressure Supervision Function

A	Pressure	D	PID Feedback (ID21)
B	Supervision Alarm Level (ID1699)	E	ON
C	PID Setpoint (ID167)	F	OFF

11.17.5.2 (ID 1699) Supervision Alarm Level

Location in the menu: **P3.15.16.2**

Use this parameter to set the overpressure limit for the Overpressure supervision. If the PID feedback becomes higher than the set overpressure limit, all auxiliary motors stop immediately. Only the regulating motor continues to operate.

11.17.6 Pump Run-time Counters

In the Multi-pump system, the time that each pump operates is monitored by a run-time counter. For example, the order that the pumps start is specified by the run-time counter values to make the wear of the pumps in the system more equal. The pump run-time counters also tell the operator to do maintenance on a pump (parameters P3.15.19.4 - P3.15.19.5).

The Pump run-time counters are in the monitoring menu, see [Table 21](#).

11.17.6.1 (ID 1673) Set Run-time Counter

Location in the menu: **P3.15.19.1**

Use this parameter to set the value that is specified by 'Set Run-time: Value' parameter to the run-time counter of the selected pump.

11.17.6.2 (ID 1087) Set Run-time Counter: Value

Location in the menu: **P3.15.19.2**

Use this parameter to set the value of the run-time counter of the selected pump when 'Set Run-time Counter' parameter is selected.

In the Multimaster and Multifollower modes, it is possible to reset or set the necessary value only to the counter Pump (1) Running Time. In the Multimaster and Multifollower modes, the monitoring value Pump (1) Running Time shows the hours of the pump that is connected to this drive. The ID number of the pump has no effect.

Example

In the Multi-pump (single drive) system, pump number 4 is replaced with a new pump. The counter value of Pump 4 Running Time must be reset.

1. Select *Pump 4* with parameter P3.15.19.3.
2. Set parameter P3.15.19.2 value to 0 h.
3. Push the button-type parameter P3.15.19.1.
4. Pump 4 Running Time is reset.

11.17.6.3 (ID 1088) Set Run-time Counter: Pump Selection

Location in the menu: **P3.15.19.3**

Use this parameter to select the pumps for which the run-time counter value is specified by 'Set Run-time: Value' parameter.

If the Multi-pump (single drive) mode is selected, the next selections are available:

- 0 = All Pumps
- 1 = Pump (1)
- 2 = Pump 2
- 3 = Pump 3
- 4 = Pump 4
- 5 = Pump 5
- 6 = Pump 6
- 7 = Pump 7
- 8 = Pump 8

If the Multifollower or Multimaster mode is selected, only the next selection is available:

- 1 = Pump (1)

In the Multimaster and Multifollower modes, it is possible to reset or to set a necessary value only for the Pump (1) Running Time. In the Multimaster and Multifollower modes, the monitoring value Pump (1) Running Time shows the hours of the pump that is connected to this drive. The ID number of the pump has no effect.

Example

In the Multi-pump (single drive) system, pump number 4 is replaced with a new pump. The counter value of Pump 4 Running Time must be reset.

1. Select *Pump 4* with parameter P3.15.19.3.
2. Set parameter P3.15.19.2 value to 0 h.
3. Push the button-type parameter P3.15.19.1.
4. Pump 4 Running Time is reset.

11.17.6.4 (ID 1109) Pump Run-time Alarm Limit

Location in the menu: **P3.15.19.4**

Use this parameter to set the alarm limit for the run-time counter of the pump. When the value of the pump run-time counter goes above this limit, a run-time counter alarm occurs.

11.17.6.5 (ID 1110) Pump Run-time Fault Limit

Location in the menu: **P3.15.19.5**

Use this parameter to set the fault limit for the run-time counter of the pump. When the value of the pump run-time counter goes above this limit, a run-time counter fault occurs.

11.17.7 Advanced Settings

11.17.7.1 (ID 15545) Staging Frequency

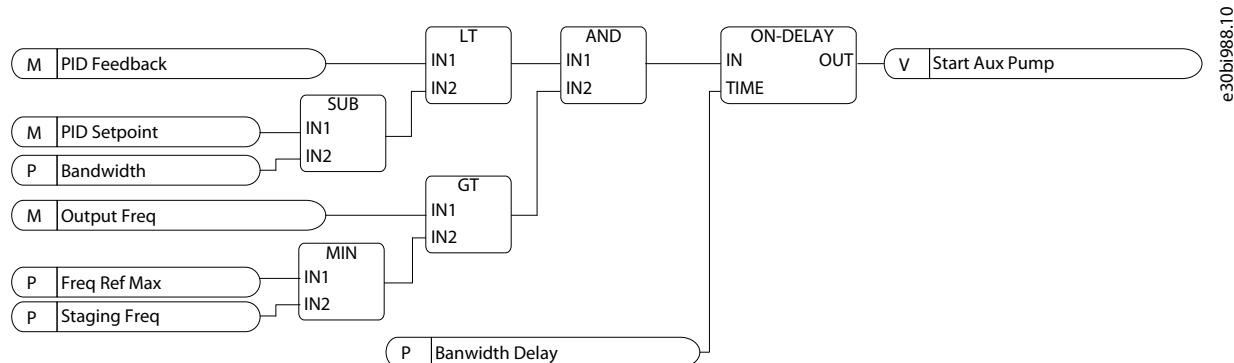
Location in the menu: **P3.15.22.1**

Use this parameter to adjust the output frequency level at which the auxiliary motor starts in the Multi-pump system.

The parameter has no effect, if its value is set above Max Frequency Reference (P3.3.1.2).

By default, an auxiliary pump starts (is staged), if the PID feedback signal goes below the specified bandwidth area and the pump that controls the system operates at the maximum frequency.

The auxiliary pump can start at a lower frequency to get better process values or to use less energy. Then, use the parameter to set the start frequency of the auxiliary pump below the maximum frequency.



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Illustration 100: Staging Frequency

11.17.7.2 (ID 15546) Destaging Frequency

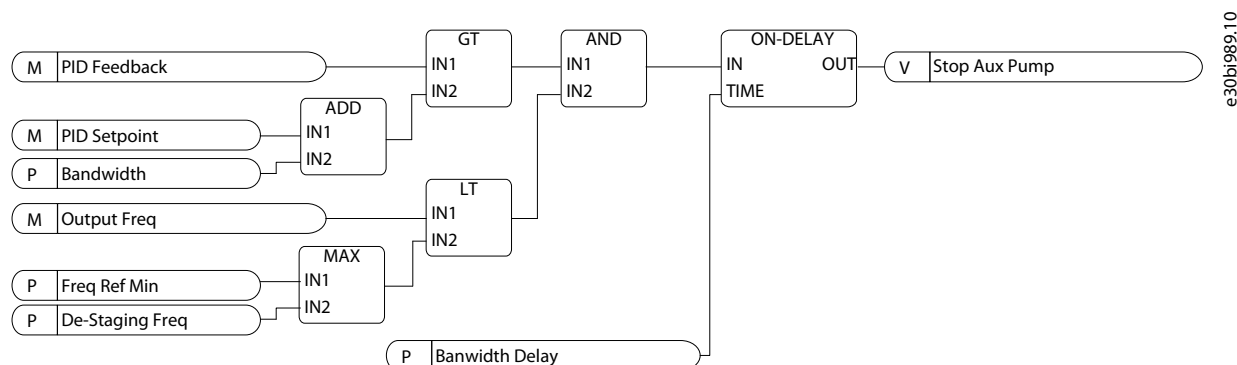
Location in the menu: **P3.15.22.2**

Use this parameter to adjust the output frequency level at which the auxiliary motor starts in the Multi-pump system.

The parameter has no effect, if its value is set below Min Frequency Reference (P3.3.1.1).

By default, an auxiliary pump stops (is destaged), if the PID feedback signal goes above the specified bandwidth area and the pump that controls the system operates at the minimum frequency.

The auxiliary pump can stop at a higher frequency to get better process values or to use less energy. Then, use the parameter to set the start frequency of the auxiliary pump above the minimum frequency.



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Illustration 101: Destaging Frequency

11.17.7.3 (ID 15588) Staging Delay

Location in the menu: **P3.15.22.3**

Use this parameter to give a delay to start the next auxiliary pump in the Multi-pump system.

When PID feedback value goes below the bandwidth area, the time defined by this parameter must elapse before the next auxiliary pump is started.

11.17.7.4 (ID 15589) Destaging Delay

Location in the menu: **P3.15.22.4**

Use this parameter to give a delay to stop auxiliary pump in the Multi-pump system.

When PID feedback value goes above the bandwidth area, the time defined by this parameter must elapse before the next auxiliary pump is stopped.

11.18 Maintenance Counters

11.18.1 The Maintenance Counters

A maintenance counter tells that maintenance must be done. For example, it is necessary to replace a belt or to replace the oil in a gearbox. There are 2 different modes for the maintenance counters, hours, or revolutions*1000. The value of the counters increases only during the RUN status of the drive.

⚠ CAUTION ⚠

MAINTENANCE HAZARD

Only an approved electrician can do maintenance. There is a risk of injury.

- Do not do maintenance if you are not approved to do it.

N O T I C E

The revolutions mode uses motor speed, which is only an estimate. The drive measures the speed every second.

When the value of a counter is more than its limit, an alarm, or a fault shows. It is possible to connect the alarm and fault signals to a digital output or a relay output.

When the maintenance is completed, reset the counter with a digital input or parameter P3.16.4 Counter 1 Reset.

11.18.2 (ID 1104) Counter 1 Mode

Location in the menu: **P3.16.1**

Use this parameter to enable the maintenance counter. A maintenance counter tells that the maintenance must be done when the counter value goes above the set limit.

11.18.3 (ID 1105) Counter 1 Alarm Limit

Location in the menu: **P3.16.2**

Use this parameter to set the alarm limit for the maintenance counter. When the value of the counter goes above this limit, a maintenance alarm occurs.

11.18.4 (ID 1106) Counter 1 Fault Limit

Location in the menu: **P3.16.3**

Use this parameter to set the fault limit for the maintenance counter. When the value of the counter goes above this limit, a maintenance fault occurs.

11.18.5 (ID 1107) Counter 1 Reset

Location in the menu: **P3.16.4**

Use this parameter to reset the maintenance counter.

11.18.6 (ID 490) Counter 1 DI Reset

Location in the menu: **P3.16.5**

Use this parameter to select the digital input that resets the value of the Maintenance Counter.

11.19 Fire Mode

11.19.1 The Fire Mode Function

When Fire mode is active, the drive resets all faults that occur and continues to operate at the same speed until it is not possible. The drive ignores all commands from the keypad, fieldbuses, and the PC tool. It only obeys the signals Fire Mode Activation, Fire Mode Reverse, Run Enable, Run Interlock 1, and Run Interlock 2 from I/O.

The Fire mode function has 2 modes, the Test mode, and the Enabled mode. To select a mode, write a password in parameter P3.17.1 (Fire Mode Password). In the Test mode, the drive does not automatically reset the faults, and the drive stops when a fault occurs.

It is also possible to configure Fire mode with the Fire mode wizard, which can be activated in the Quick Setup menu with parameter B1.1.4.

When the Fire mode function is activated, an alarm shows on the display.

N O T I C E

WARRANTY BECOMING VOID

The warranty is void if the Fire mode function is activated.

- Use Test mode to test the Fire mode function and the warranty stays valid.

11.19.2 (ID 1599) Fire Mode Password

Location in the menu: **P3.17.1**

Use this parameter to enable the fire mode function.

N O T I C E

All other fire mode parameters are locked when the fire mode is enabled and correct password is set in this parameter.

Selection number	Selection name	Description
1002	Enabled mode	The drive resets all the faults and continues to operate at the same speed until it is not possible.
1234	Test mode	The drive does not automatically reset the faults, and the drive stops when a fault occurs.

11.19.3 (ID 1617) Fire Mode Frequency Source

Location in the menu: **P3.17.2**

Use this parameter to select the frequency reference source when the fire mode is active. This parameter enables the selection of, for example, the AI1, or the PID controller as the reference source when operating the fire mode.

11.19.4 (ID 1598) Fire Mode Frequency

Location in the menu: **P3.17.3**

Use this parameter to set the frequency that is used when Fire mode is active. The drive uses this frequency when the value of parameter P3.17.2 Fire Mode Frequency Source is Fire Mode Frequency.

11.19.5 (ID 1596) Fire Mode Activation on Open

Location in the menu: **P3.17.4**

Use this parameter to select the digital input signal that activates the Fire Mode function. If this digital input signal is activated, an alarm shows on the display, and the warranty becomes void. The type of this digital input signal is NC (normally closed).

It is possible to try the Fire mode with the password that activates the Test mode. Then the warranty stays valid.

N O T I C E

If Fire mode is enabled, and the correct password is given to the parameter Fire Mode Password, all the Fire mode parameters become locked. To change the Fire mode parameters, change the value of P3.17.1 Fire Mode Password to 0 first.

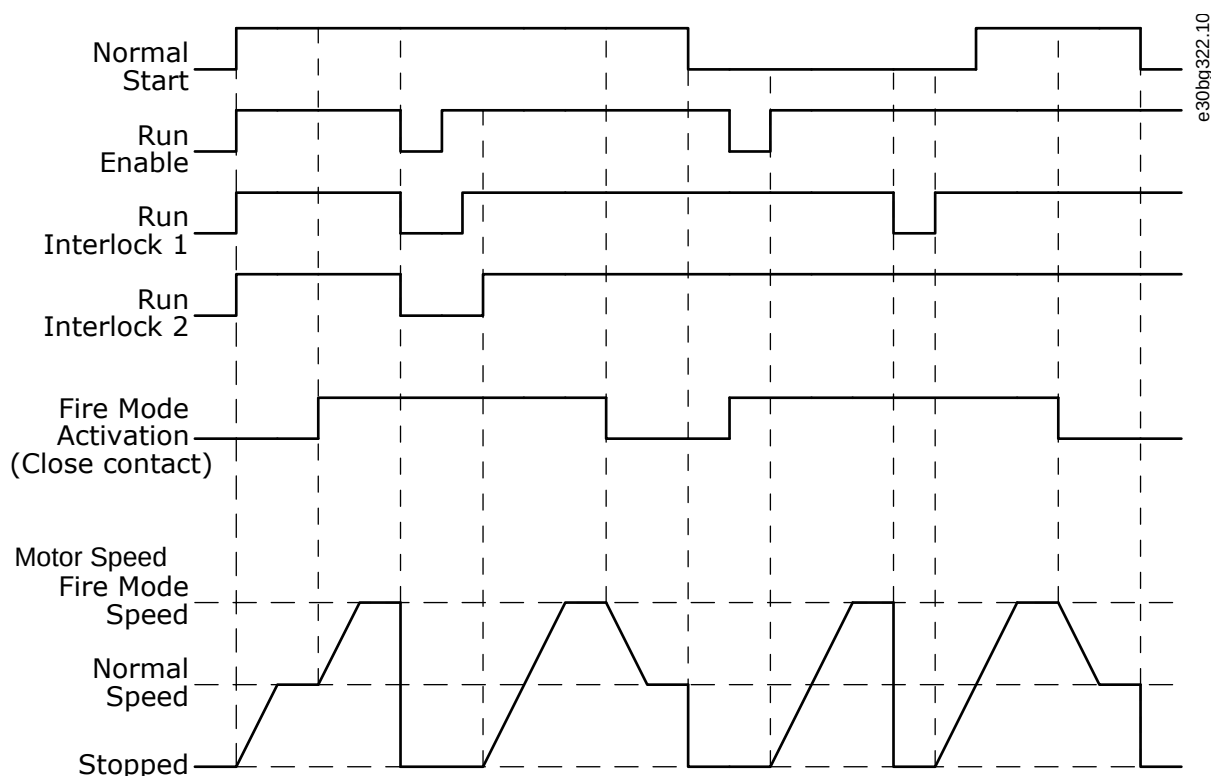


Illustration 102: The Fire Mode Function

11.19.6 (ID 1619) Fire Mode Activation on Close

Location in the menu: **P3.17.5**

Use this parameter to select the digital input signal that activates the Fire Mode function. The type of this digital input signal is NO (normally open). See the description for P3.17.4 Fire Mode Activation on Open.

11.19.7 (ID 1618) Fire Mode Reverse

Location in the menu: **P3.17.6**

Use this parameter to select the digital input signal that gives a command for reverse rotation direction during the Fire Mode. The parameter does not affect normal operation.

If it is necessary for the motor to operate always FORWARD or always REVERSE in Fire Mode, select the correct digital input.

DigIn Slot0.1 = always FORWARD

DigIn Slot0.2 = always REVERSE

11.19.8 (ID 1597) Fire Mode Status

Location in the menu: **V3.17.7**

This monitoring value shows the status of the fire mode function.

11.19.9 (ID 1679) Fire Mode Counter

Location in the menu: **V3.17.8**

This monitoring value shows the number of the fire mode activations.

The counter cannot be reset.

11.19.10 (ID 15580) Fire Mode Run Indication Current

Location in the menu: **P3.17.9**

Use this parameter to set the current limit for the run indication signal of the relay output.

This parameter only affects if the value "Run indication" is selected for a relay output and the fire mode is active. The "Run indication" relay output function tells quickly if current is supplied to the motor during a fire.

The value of this parameter is the percentage counted from the motor nominal current. If there is a fire and the current that is supplied to the motor is more than the nominal current times the value of this parameter, the relay output closes.

For example, if parameter Motor Nominal Current has the value 5 A, and 20% is set as the default value of this parameter, the relay output closes and Fire Mode activates when the output current goes to 1 A.

This parameter does not affect if the fire mode is not active. In normal operation, if "Run indication" is selected for a relay output, the result is the same as when "Run" is selected for the relay output.

11.20 Motor Preheat Function

11.20.1 (ID 1225) Motor Preheat Function

Location in the menu: **P3.18.1**

Use this parameter to enable or disable the Motor Preheat function. The motor preheat function keeps the drive and the motor warm during the STOP status. In the motor preheat, the system gives the motor a DC current. The motor preheat prevents, for example, condensation.

Selection number	Selection name	Description
0	Not used	The motor preheat function is disabled.
1	Always in Stop state	The motor preheat function is activated always when the drive is in the Stop state.
2	Controlled by digital input	The motor preheat function is activated by a digital input signal, when the drive is in the Stop state. It is possible to make the selection of the digital input for the activation with parameter P3.5.1.18.
3	Temperature limit (heat sink)	The motor preheat function is activated if the drive is in the Stop state, and the temperature of the heat sink of the drive goes below the temperature limit that was set with parameter P3.18.2.
4	Temperature limit (measured motor temperature)	The motor preheat function is activated if the drive is in the Stop state, and the measured motor temperature goes below the temperature limit that was set with parameter P3.18.2. It is possible to set the measurement signal of the motor temperature with parameter P3.18.5. NOTE! To use this operation mode, an option board for temperature measurement (for example OPTBH) is needed.

11.20.2 (ID 1226) Preheat Temperature Limit

Location in the menu: **P3.18.2**

Use this parameter to set the temperature limit of the Motor Preheat function. The motor preheat becomes active when the heat sink temperature or the measured motor temperature goes below this level, and when P3.18.1 is set to 3 or 4.

11.20.3 (ID 1227) Motor Preheat Current

Location in the menu: **P3.18.3**

Use this parameter to set the DC current of the Motor Preheat function. The DC current for the pre-heating of the motor and the drive in stop state. Activated as in P3.18.1.

11.20.4 (ID 1044) Motor Preheat ON

Location in the menu: **P3.18.4**

Use this parameter to select the digital input signal that activates Motor Preheat function. This parameter is used when P3.18.1 is set to 2. When the value for P3.18.1 is 2, it is also possible to connect time channels to this parameter.

11.20.5 (ID 1045) Preheat Motor Temperature

Location in the menu: **P3.18.5**

Use this parameter to select the temperature signal that is used to measure the motor temperature for motor preheat function.

NOTICE

This parameter is not available if there is no option board for temperature measurement.

11.21 Drive Customizer

11.21.1 (ID 15001) Operation Mode

Location in the menu: **P3.19.1**

Use this parameter to select the operation mode of the drive customizer.

Selection number	Selection name	Description
0	Execute Program	The Drive customizer is running. Configuration is not allowed for the Drive customizer.
1	Programming	The Drive customizer is not running. Configuration is allowed for the Drive customizer.

11.22 Pump Control

11.22.1 Auto-Cleaning

Use the Auto-cleaning function to remove dirt or other material from the pump impeller. It is also possible to use the function to clear a blocked pipe or valve. It is possible to use the auto-cleaning, for example, in wastewater systems to keep the performance of the pump satisfactory.

11.22.1.1 (ID 1714) Cleaning Function

Location in the menu: **P3.21.1.1**

Use this parameter to enable the Auto-cleaning function.

Selection number	Selection name	Description
0	Disabled	
1	Enabled (DIN)	The cleaning sequence is started with a digital input signal. If the start command of the drive is active, a rising edge of the digital input signal (P3.21.1.2) starts the cleaning sequence. If the drive is in the Sleep mode (PID sleep), the cleaning sequence can also be activated.
2	Enabled (Current)	The cleaning sequence starts when the motor current goes above the current limit (P3.21.1.3) for a longer time than specified by P3.21.1.4.
3	Enabled (Real Time)	The cleaning sequence agrees with the internal real-time clock of the drive.

N O T I C E

A battery must be installed in the real-time clock.

If the start command of the drive is active, the cleaning sequence starts on the selected weekdays (P3.21.1.5) at the specified time of the day (P3.21.1.6). If the drive is in the Sleep mode (PID sleep), the cleaning sequence can also be activated.

To stop the cleaning sequence, deactivate the start command of the drive. When 0 is selected, the cleaning function is not used.

11.22.1.2 (ID 1715) Cleaning Activation

Location in the menu: **P3.21.1.2**

Use this parameter to select the digital input signal that starts the Auto-cleaning sequence. The auto-cleaning stops if the activation signal is removed before the sequence is complete.

N O T I C E

If the input is activated, the drive starts.

11.22.1.3 (ID 1712) Cleaning Current Limit

Location in the menu: **P3.21.1.3**

Use this parameter to set the current limit at which the Auto-cleaning starts. If the current of the motor stays above this limit for longer than the set time, an autocleaning sequence starts.

11.22.1.4 (ID 1713) Cleaning Current Delay

Location in the menu: **P3.21.1.4**

Use this parameter to set the time that the motor current must stay above the limit before the auto-cleaning starts.

Parameters P3.21.1.3 and P3.21.1.4 are used only when P3.21.1.1 = 2.

The cleaning sequence starts when the motor current goes above the current limit (P3.21.1.3) for longer than specified with P3.21.1.4. The current limit is specified as a percentage of the motor nominal current.

11.22.1.5 (ID 1715) Cleaning Weekdays

Location in the menu: **P3.21.1.5**

Use this parameter to set the days of the week when the Auto-cleaning is executed. This parameter is used only when P3.21.1.1 = 3.

11.22.1.6 (ID 1700) Cleaning Time of Day

Location in the menu: **P3.21.1.6**

Use this parameter to set the time of the day when the Auto-cleaning is executed. This parameter is used only when P3.21.1.1 = 3.

N O T I C E

A battery must be installed in the real-time clock.

11.22.1.7 (ID 1716) Cleaning Cycles

Location in the menu: **P3.21.1.7**

Use this parameter to set the number of forward or reverse cleaning cycles.

11.22.1.8 (ID 1717) Clean Forward Frequency

Location in the menu: **P3.21.1.8**

Use this parameter to set the frequency reference of the drive for the forward direction in the Auto-cleaning cycle.

Set the frequency and time of the cleaning cycle with the parameters P3.21.1.4, P3.21.1.5, P3.21.1.6, and P3.21.1.7.

11.22.1.9 (ID 1718) Clean Forward Time

Location in the menu: **P3.21.1.9**

Use this parameter to set the operation time for the forward direction frequency in the Autocleaning cycle. See parameter [11.22.1.8 \(ID 1717\) Clean Forward Frequency](#).

11.22.1.10 (ID 1719) Clean Reverse Frequency

Location in the menu: **P3.21.1.10**

Use this parameter to set the frequency reference of the drive for the reverse direction in the Auto-cleaning cycle. See parameter [11.22.1.8 \(ID 1717\) Clean Forward Frequency](#).

11.22.1.11 (ID 1720) Clean Reverse Time

Location in the menu: **P3.21.1.11**

Use this parameter to set the operation time for the reverse direction frequency in the Autocleaning cycle. See parameter [11.22.1.8 \(ID 1717\) Clean Forward Frequency](#).

11.22.1.12 (ID 1721) Cleaning Acceleration Time

Location in the menu: **P3.21.1.12**

Use this parameter to set the motor acceleration time when the Auto-cleaning is active. It is possible to set acceleration and deceleration ramps for the Auto-cleaning function with parameters P3.21.1.12 and P3.21.1.13.

11.22.1.13 (ID 1722) Cleaning Deceleration Time

Location in the menu: **P3.21.1.13**

Use this parameter to set the motor deceleration time when the Auto-cleaning is active. It is possible to set acceleration and deceleration ramps for the Auto-cleaning function with parameters P3.21.1.12 and P3.21.1.13.

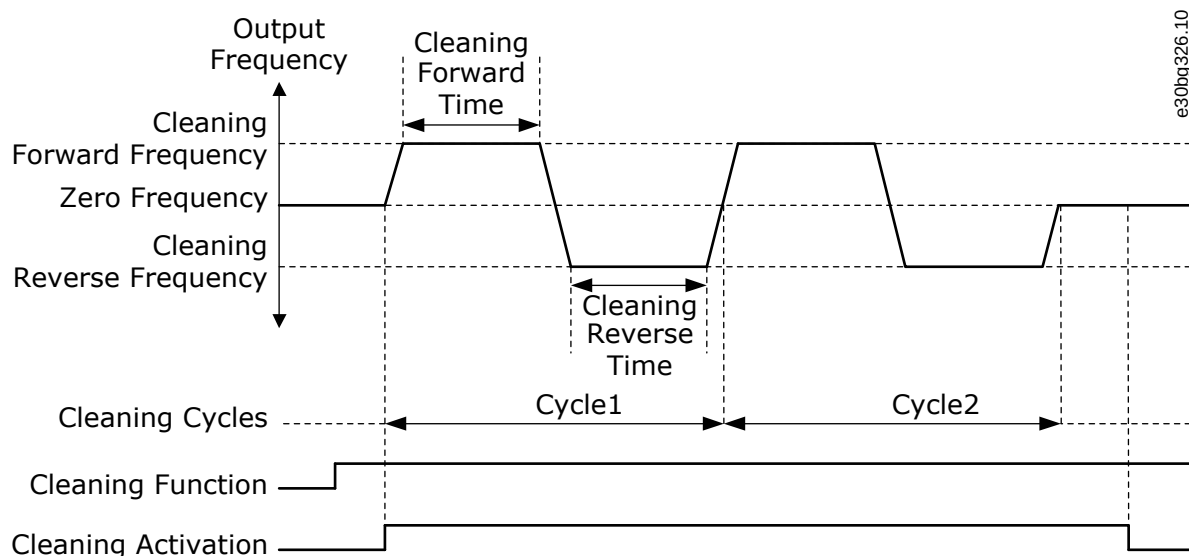


Illustration 103: The Auto-cleaning Function

11.22.2 Jockey Pump

11.22.2.1 (ID 1674) Jockey Function

Location in the menu: **P3.21.2.1**

Use this parameter to control the Jockey pump function. A Jockey pump is a smaller pump that keeps the pressure in the pipeline, when the main pump is in the sleep mode. It can occur, for example, in the night.

The Jockey pump function controls a jockey pump with a digital output signal. A jockey pump can be used if a PID controller is used to control the main pump. The function has 3 operation modes.

Selection number	Selection name	Description
0	Not used	
1	PID sleep	The jockey pump starts when the PID Sleep of the main pump activates. The jockey pump stops when the main pump wakes up from the sleep mode.
2	PID sleep (level)	The jockey pump starts when the PID Sleep activates, and the PID feedback signal is less than the level set by parameter P3.21.2.2. The jockey pump stops when the PID feedback signal is more than the level set in parameter P3.21.2.3 or the main pump wakes up from the sleep mode.

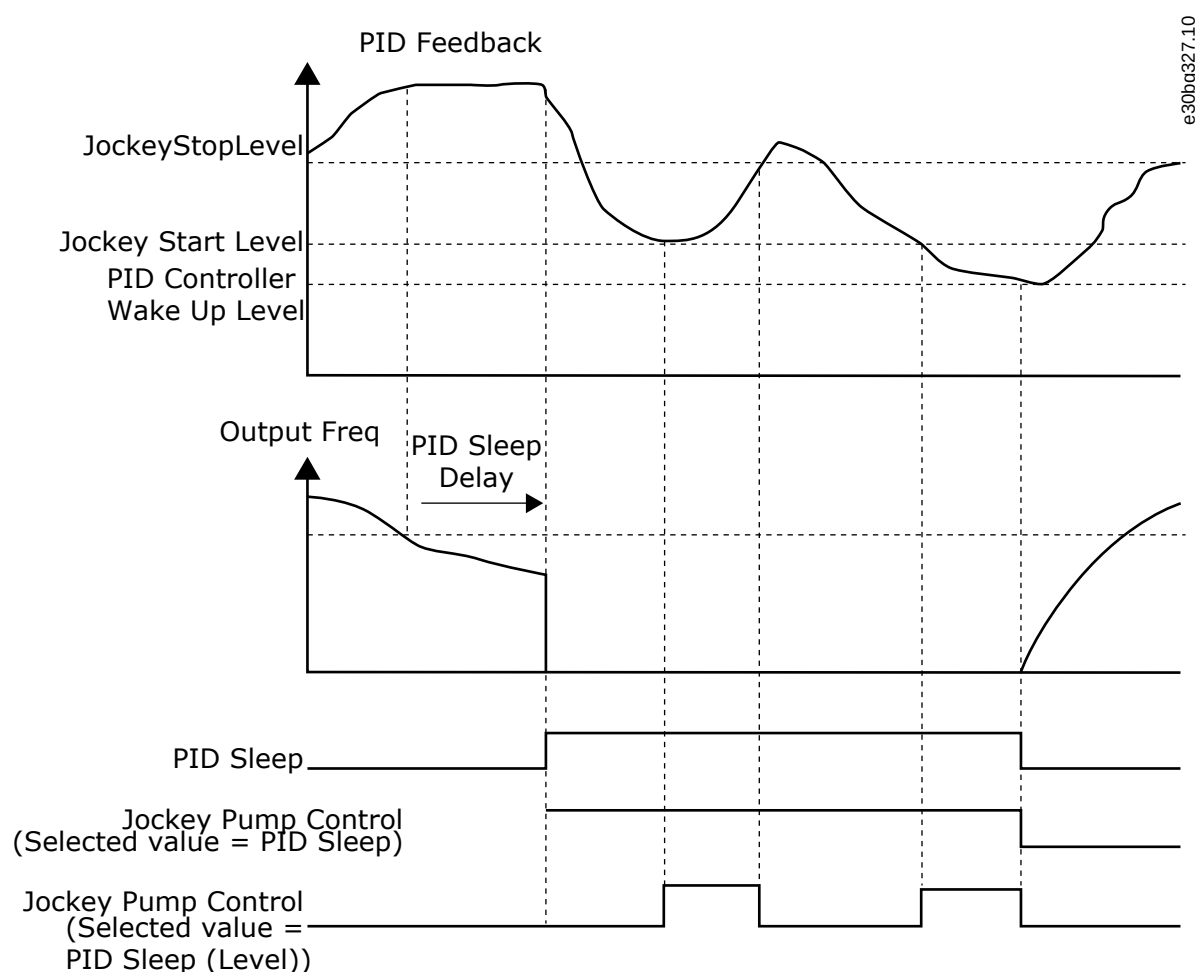


Illustration 104: The Jockey Pump Function

11.22.2.2 (ID 1675) Jockey Start Level

Location in the menu: **P3.21.2.2**

Use this parameter to set the level of the PID feedback signal at which the jockey pump starts when the main pump is in the sleep state. The jockey pump starts when PID Sleep is active and the PID feedback signal goes below the level set in this parameter.

This parameter is used only if P3.21.2.1 = 2 PID sleep (level).

11.22.2.3 (ID 1676) Jockey Stop Level

Location in the menu: **P3.21.2.3**

Use this parameter to set the level of the PID feedback signal at which the jockey pump stops when the main pump is in the sleep state.

The jockey pump stops when PID Sleep is active and the PID feedback signal goes above the level set in this parameter, or when the PID controller wakes up from sleep mode.

This parameter is used only if P3.21.2.1 = 2 PID sleep (level).

11.22.3 Priming Pump

A priming pump is a smaller pump that primes the inlet of the main pump to prevent suction of air.

The priming pump function controls a priming pump with a digital output signal. It is possible to set a delay to start the priming pump before the main pump starts. The priming pump operates continuously while the main pump operates. If the main pump goes into sleep mode, the priming pump also stops for that time. When waking up from sleep mode, the main pump, and the priming pump start simultaneously.

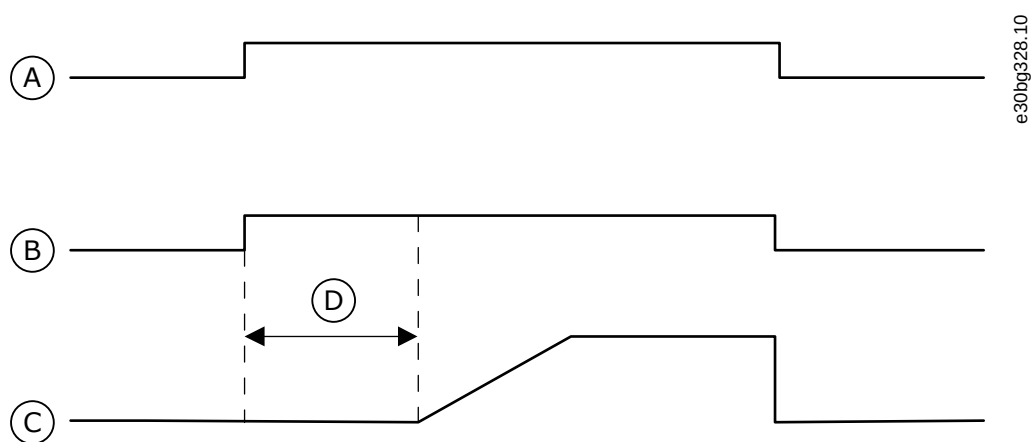


Illustration 105: The Priming Pump Function

A	Start Command (Main Pump)	C	Output Frequency (Main Pump)
B	Priming Pump Control (Digital Output Signal)	D	Priming Time

11.22.3.1 (ID 1677) Priming Function

Location in the menu: **P3.21.3.1**

Use this parameter to enable the Priming pump function. A priming pump is a smaller pump that primes the inlet of the main pump to prevent the suction of air. The priming pump function controls a priming pump with a relay output signal.

11.22.3.2 (ID 1678) Priming Time

Location in the menu: **P3.21.3.2**

Use this parameter to set the time that the priming pump operates before the main pump starts.

11.22.4 Anti-blocking Function

The Anti-blocking function makes the pump not to get blocked if the pump is stopped in the Sleep mode for a long time. The pump starts at intervals, while it is in the Sleep mode. It is possible to make a configuration of the interval, run-time, and speed for the anti-blocking.

11.22.4.1 (ID 1696) Anti-blocking Interval

Location in the menu: **P3.21.4.1**

Use this parameter to set the interval for the Anti-blocking function. This parameter gives the time after which the pump starts at the specified speed (P3.21.4.3 Anti-blocking Frequency) and for the specified quantity time (P3.21.4.2 Anti-blocking Run-time).

The Anti-blocking function can be used in the Single drive and the Multidrive systems only when the pump is in the sleep mode, or in the standby mode (Multidrive system).

The Anti-blocking function is enabled when the value of this parameter is more than 0 and disabled when the value is 0.

11.22.4.2 (ID 1697) Anti-blocking Run-time

Location in the menu: **P3.21.4.2**

Use this parameter to set the time that the pump operates at the set speed when the Antiblocking function is activated.

11.22.4.3 (ID 1504) Anti-blocking Frequency

Location in the menu: **P3.21.4.3**

Use this parameter to set the frequency reference of the drive that is used when the antiblocking function is activated.

11.22.5 Frost Protection

Use the Frost protection function to protect the pump from frost damages. If the pump is in sleep mode and the temperature that is measured in the pump goes below the set protection temperature, operate the pump at a constant frequency. Constant frequency

is set in P3.13.10.6 Frost Protection Frequency. To use the function, a temperature transducer, or a temperature sensor must be installed on the pump covering or on the pipe line near the pump.

11.22.5.1 (ID 1704) Frost Protection

Location in the menu: **P3.21.5.1**

Use this parameter to enable the Frost Protection function. If the measured temperature of the pump goes below the set level and the drive is in sleep state, the frost protection makes the pump start and operate at a constant frequency.

11.22.5.2 (ID 1705) Temperature Signal

Location in the menu: **P3.21.5.2**

Use this parameter to select the source of the temperature signal that is used for the Frost Protection function.

11.22.5.3 (ID 1706) Temperature Signal Minimum

Location in the menu: **P3.21.5.3**

Use this parameter to set the minimum value of the temperature signal. For example, a temperature signal range of 4...20 mA agrees to the temperature of -50...200 °C.

11.22.5.4 (ID 1707) Temperature Signal Maximum

Location in the menu: **P3.21.5.4**

Use this parameter to set the maximum value of the temperature signal. For example, a temperature signal range of 4...20 mA agrees to the temperature of -50...200 °C.

11.22.5.5 (ID 1708) Frost Protection Temperature Limit

Location in the menu: **P3.21.5.5**

Use this parameter to set the temperature limit at which the drive starts. If the temperature of the pump goes below this limit and the drive is in the sleep state, the frost protection function starts the drive.

11.22.5.6 (ID 1710) Frost Protection Frequency

Location in the menu: **P3.21.5.6**

Use this parameter to set the frequency reference of the drive that is used when the frost protection function is activated.

11.22.5.7 (ID 1711) Frost Temperature Monitoring

Location in the menu: **P3.21.5.7**

This monitoring value shows the value of the temperature signal that is used for Frost Protection function.

11.22.6 Flow Switch

11.22.6.1 (ID 15590) Flow Switch

Location in the menu: **P3.21.6.1**

Use this parameter to select the digital input signal that is the feedback signal from the Flow Switch.

11.22.6.2 (ID 15585) Flow Switch Fault Delay

Location in the menu: **P3.21.6.2**

Use this parameter to set the time that the feedback signal from Flow Switch must stay inactive before the Flow Switch fault occurs.

11.22.6.3 (ID 15586) Flow Switch Bypass at Start

Location in the menu: **P3.21.6.3**

Use this parameter to set the time after which the Flow Switch supervision becomes active at the start-up of the drive.

11.22.6.4 (ID 15587) Flow Switch Minimum Frequency

Location in the menu: **P3.21.6.4**

Use this parameter to set the output frequency level below which the Flow Switch supervision is disabled.

11.23 Advanced Harmonic Filter

11.23.1 (ID 15510) Capacitor Disconnection Limit

Use this parameter to set the disconnection limit for the advanced harmonic filter. The value is in percentage of the drive nominal power.

11.23.2 (ID 15511) Capacitor Disconnection Hysteresis

Use this parameter to set the disconnection hysteresis for the advanced harmonic filter. The value is in percentage of the drive nominal power.

11.23.3 (ID 15513) AHF Over Temperature

- Location in the menu: **P3.5.1.59** (In I/O Configuration parameter menu)
- Location in the menu: **P3.23.3** (In Advanced Harmonic Filter parameter menu)

Use this parameter to set the digital input signal that activates AHF Over Temp (fault ID 1118).

11.23.4 (ID 15512) AHF Fault Response

Use this parameter to select the response of the AC drive to an AHF Over Temp fault.

11.24 Condition-based Monitoring

11.24.1 Introduction to Condition-based Monitoring

Condition-based monitoring in VACON® 100 drives is launched as a factory-flashed licensed firmware, embedded within the drive. Benefits of installing the condition-based monitoring firmware are as follows:

- Reduces unexpected downtime.
- Optimizes drive or motor working conditions.
- Eliminates unexpected halts in production.

Condition-based monitoring enables regularly checks of the condition and performance of the machine when the drive is in service and detects mechanical, motor, or application failures in advance. Corrective actions can be performed before the process or application is impacted. Alarm/fault or warnings are triggered in the drive to notify customers or service technicians. Some of the corrective actions include replacement of faulty motors or bearings and ensuring the motor is running within optimal conditions.

Following are the monitoring capabilities introduced:

- **Motor stator winding monitoring:** During monitoring, inter-turn short circuit or unbalance in the motor winding is detected in advance. Damage caused by motor stator winding isolation occurs over time. When more winding turns are impacted, the over-current protection is activated due to extensive heating and stops the motor.
- **Vibration monitoring:** With the help of external sensors, the drive can monitor vibration levels in a motor. Vibrations affect motor control and can lead to motor failure. During monitoring, early detection of motor misalignment is detected and wear and tear of mechanical parts are identified earlier.

NOTE! ISO 10816 standard provides guidance for evaluating vibration severity for machines operating within 10–200 Hz of frequency range. The standard must be complied with before commissioning of vibration monitoring function.

- **Load envelope:** Mechanical load of an application is monitored by comparing current load curve with expected load curve based on data gathered during commissioning. During monitoring, overload and underload deviations which occur in applications are detected.

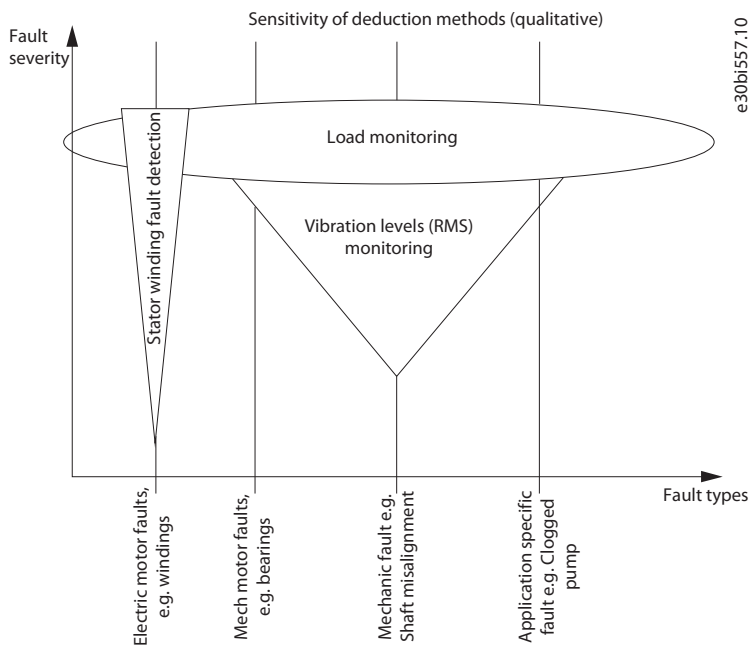


Illustration 106: Condition-based Monitoring

11.24.1.1 Stages of Condition-based Monitoring

The following figure describes the different stages of Condition-based Monitoring.

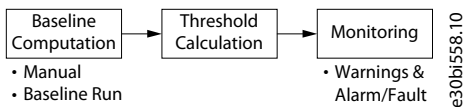


Illustration 107: Stages of Condition-based Monitoring

- **Baseline Computation:** During the initial stage, baselines for different types of condition-based monitoring are computed, based on the type of baseline mode selected.
- **Threshold Calculation:** Once the baseline is computed, the thresholds for warnings (stage 1 and stage 2) and alarm/fault are computed based on the warning or alarm/fault mode. The threshold and mode are set via parameters.
- **Monitoring:** After the thresholds are calculated, condition-based monitoring begins. Alarm/fault and warnings are triggered during deviations to notify customers.

To begin condition-based monitoring, a baseline must be generated. During this activity, the system captures motor stator winding speed points for each baseline. Define the duration, minimum and maximum speed for baseline generation. During baseline generation, 10 speed points are captured. The first and last speed point is captured based on the defined minimum and maximum speed respectively.

When speed points are not captured properly, baseline generation fails, then a new baseline has to be generated. The baseline serves as a reference for threshold limits. Using parameters, select the type of baseline measurement.

- **Baseline Run:** The drive controls the motor speed and monitors required values to derive a baseline. The drive creates a speed profile for baseline measurement. The drive interrupts the application during baseline run to operate on the speed profile created for baseline measurement. The duration for baseline calculation can be specified. Baseline measurements can copy to other drive or return to same drive after factory reset.
- **Manual Run:** The measurement points can be manually configured directly through modified parameters. Manual run can be made before or after baseline run.

11.24.2 (ID 3501) BaselineRun Start

Location in the menu: **P3.24.2.1**

Use this parameter to activate baseline run.

Baseline run is made before normal operation. Idea is to compare actual measurements with measurements taken when the system was in healthy/normal condition. The measurements from healthy condition are called "baseline".

In baseline run, the measurements of the motor current unbalance for stator winding, vibration, and motor torque are collected.

The baseline run goes through 10 frequency points including 10 steady and 9 ramp states. It collects the minimum, maximum, mean, and standard deviation values, and stores the statistical baseline data in arrays. Normal start command is required after baseline run start.

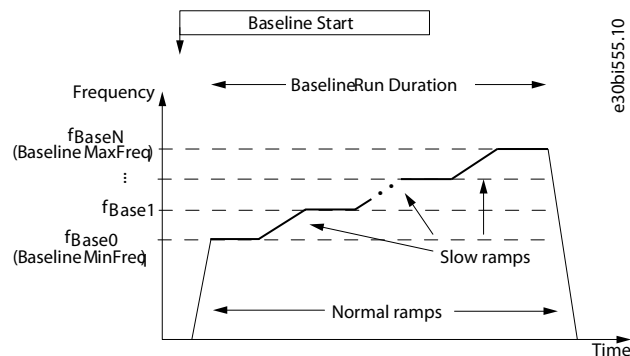


Illustration 108: BaselineRun Start

11.24.3 (ID 3502) BaselineRun MinFreq

Location in the menu: **P3.24.2.2**

Use this parameter to set frequency point where baseline run is started.

11.24.4 (ID 3503) BaselineRun MaxFreq

Location in the menu: **P3.24.2.3**

Use this parameter to set frequency point where baseline run is ended.

11.24.5 (ID 3504) BaselineRun Duration

Location in the menu: **P3.24.2.4**

Use this parameter to set the duration of the baseline run.

11.24.6 Manual Run

Before or after baseline run, each measurement point can be modified by parameter. One array includes 10 points in steady state and 9 points in ramp state.

11.24.6.1 (ID 3506) Modified Array

Location in the menu: **P3.24.2.5**

Use this parameter to select a modified array.

11.24.6.2 (ID 3507) Modified Point

Location in the menu: **P3.24.2.6**

Use this parameter to select which measurement point to modify in the array. Steady array includes 10 points and ramp array includes 9 points. First point has value 0.

11.24.6.3 (ID 3508) Modified Value

Location in the menu: **P3.24.2.7**

Use this parameter to give new value to the point selected with *ID 3507*.

11.24.6.4 (ID 3505) Modified Activation

Location in the menu: **P3.24.2.8**

Before or after the baseline run, each measurement point can be modified. Use this parameter to activate the *Modified value (ID 3508)* of the *Modified point (ID 3507)*.

11.24.6.5 (ID 3509) Baseline Data Selector

Location in the menu: **P3.24.2.9**

Use this parameter to select which array points can be monitored in monitor group *M2.16.1 Baseline data*.

11.24.6.6 (ID 3540) Condition-based Monitoring Fault Mode

Location in the menu: **P3.24.1**

Use this parameter to set the response of the drive to Exception.

Selection number	Selection name	Description
0	No Action	
1	Warnings	Warning levels S1 and S2 are used.
2	Fault + Warnings	Warning S1, warning S2 and alarm/fault levels are used.

11.24.7 Stator Winding Monitoring

For stator winding monitoring, the motor currents are being evaluated for evolving unbalances. For this purpose, current unbalance is computed and monitored. If there are the unbalanced grid voltages, a resonance-like oscillation can occur if the motor frequency matches the grid frequency.

11.24.7.1 (ID 1913) Line Freq

Location in the menu: **P3.24.3.1**

Use this parameter to set the grid Nominal Frequency.

11.24.7.2 (ID 1914) Line Freq Hysteresis

Location in the menu: **P3.24.3.2**

Use this parameter to set the hysteresis of the grid Nominal Frequency. Stator winding monitoring is not used in the hysteresis area.

11.24.8 Threshold Value

The several statistical data points (mean, minimum, maximum, and standard) from the baseline model can be combined one threshold value.

Threshold value = MeanFactor * mean + MinFactor * min + MaxFactor * max + StdFactor * stddev

11.24.8.1 (ID 3511) Current Mean Factor

Location in the menu: **P3.24.3.3.1.1**

Use this parameter to set the mean factor in the threshold value. If the threshold value is zero, the mean factor is not used.

11.24.8.2 (ID 3512) Current Min Factor

Location in the menu: **P3.24.3.3.1.2**

Use this parameter to set min factor in the threshold value. If the threshold value is zero, the min factor is not used.

11.24.8.3 (ID 3513) Current Max Factor

Location in the menu: **P3.24.3.3.1.3**

Use this parameter to set the max factor in the threshold value. If the threshold value is zero, the max factor is not used.

11.24.8.4 (ID 3514) Current Std Factor

Location in the menu: **P3.24.3.3.1.4**

Use this parameter to set the std factor in the threshold value. If the threshold value is zero, the std factor is not used.

11.24.8.5 (ID 3515) Current Interpolation Type

Location in the menu: **P3.24.3.3.1.5**

Use this parameter to select interpolation type for the notification threshold values across frequency points.

11.24.8.6 (ID 3550) Vibration Mean Factor

Location in the menu: **P3.24.4.2.1**

Use this parameter to set the mean factor in the threshold value. If the threshold value is zero, the mean factor is not used.